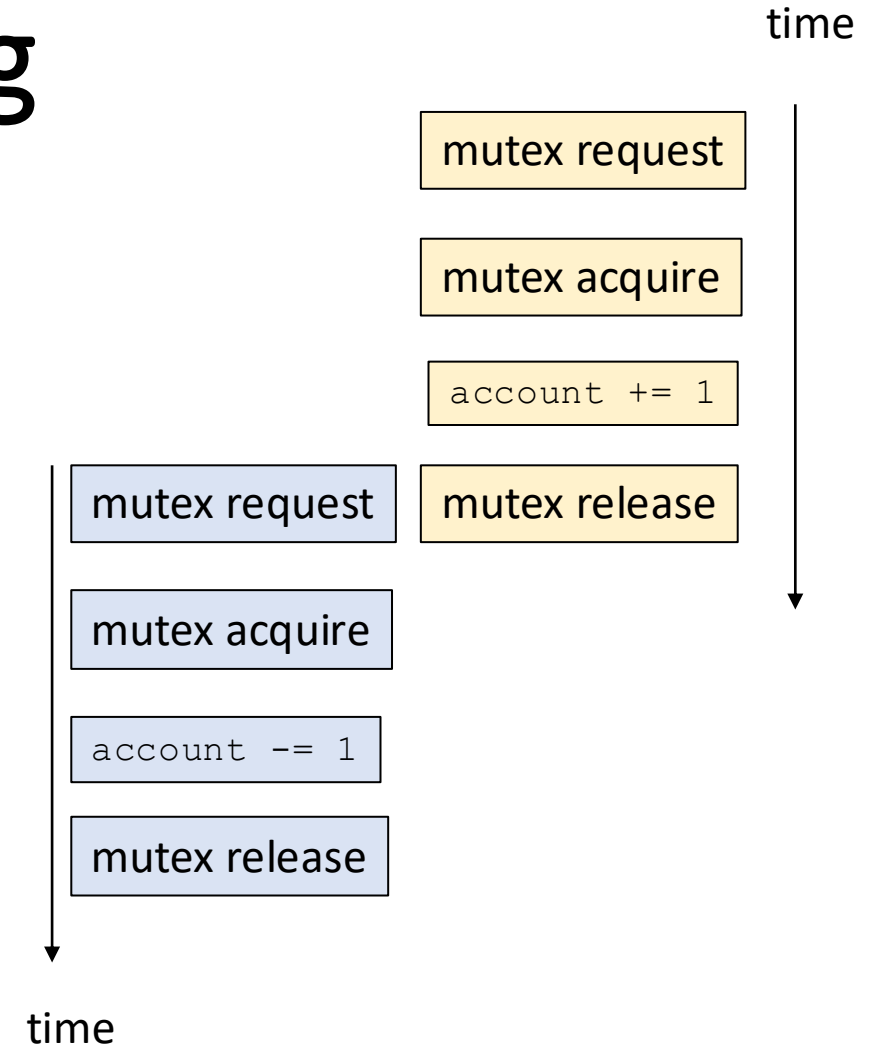


CSE113: Parallel Programming

- **Topics:**

- Intro to mutual exclusion
 - Different types of parallelism
 - Data conflicts
 - Protecting shared data



Announcements

- Homework due on Oct 8
 - You have everything you need to get it done
 - Three free late days, nothing accepted after that
 - Plenty of office hours remaining to get help
 - Work on your design before asking for help
- Starting on Module 2 today!

Announcements

- Homework 1 notes:
 - No assigned speedup required. You should get a noticeable speedup from ILP
 - You can start to share results on your personal machines. Everyone's results will be slightly different
 - Sometimes you cannot account for small differences
 - Run your code for more iterations and take an average

Previous quiz

Previous quiz

How many elements of type double can be stored in a cache line?

☐ 1

☐ 2

☐ 4

☐ 8

☐ 16

☐ 32

double is often 8 bytes.

Cache line size: Intel: 64 bytes. Apples: 128 bytes.

$64 / 8 = 8$, $128 / 8 = 16$

Previous quiz

Instructions with the following property should be placed as far apart as possible in machine code:

-
- ☐ Instructions that compute floating point values
 - ☐ Instructions that load from memory
 - ☐ Instructions that depend on each other
 - ☐ Instructions that perform the same operation

Previous quiz

What does ILP stand for?

☐ Interleaved Language Program

☐ Instruction Level Parallelism

☐ Interpreted Latency Pipeline

Previous quiz

C++ threads are initialized with a function argument where they will start execution, but they must be explicitly started with the "launch" command.

☐ True

☐ False

Previous quiz

The "join" function for a C++ thread causes the thread to immediately exit.

☐ True

☐ False

Previous quiz

A thread that is launched will eventually exit by itself and there is no need for the main thread to keep track of the threads it launches.

☐ True

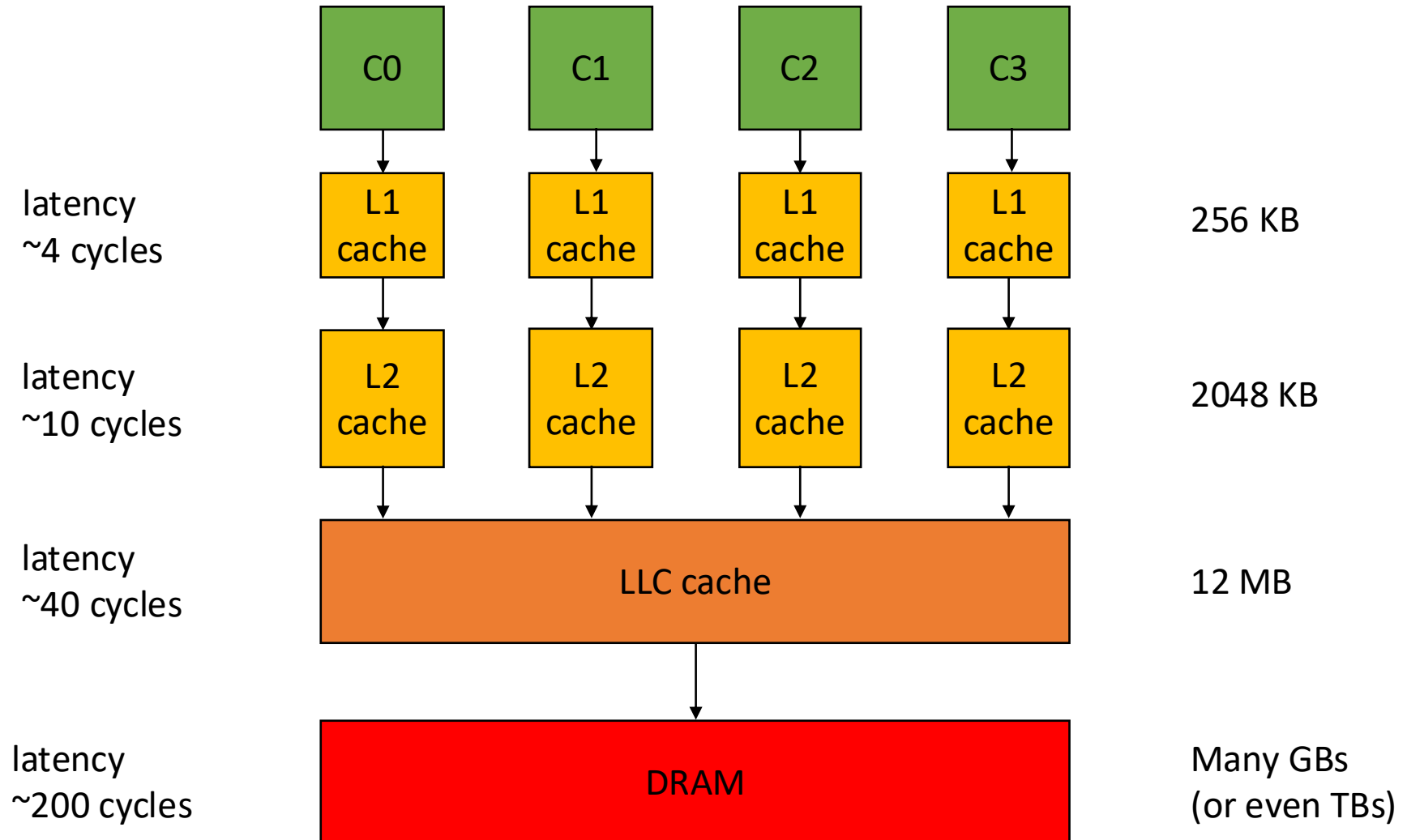
☐ False

Previous quiz

In 2 or 3 sentences, explain the difference between instruction level parallelism and thread parallelism

Review

Caches

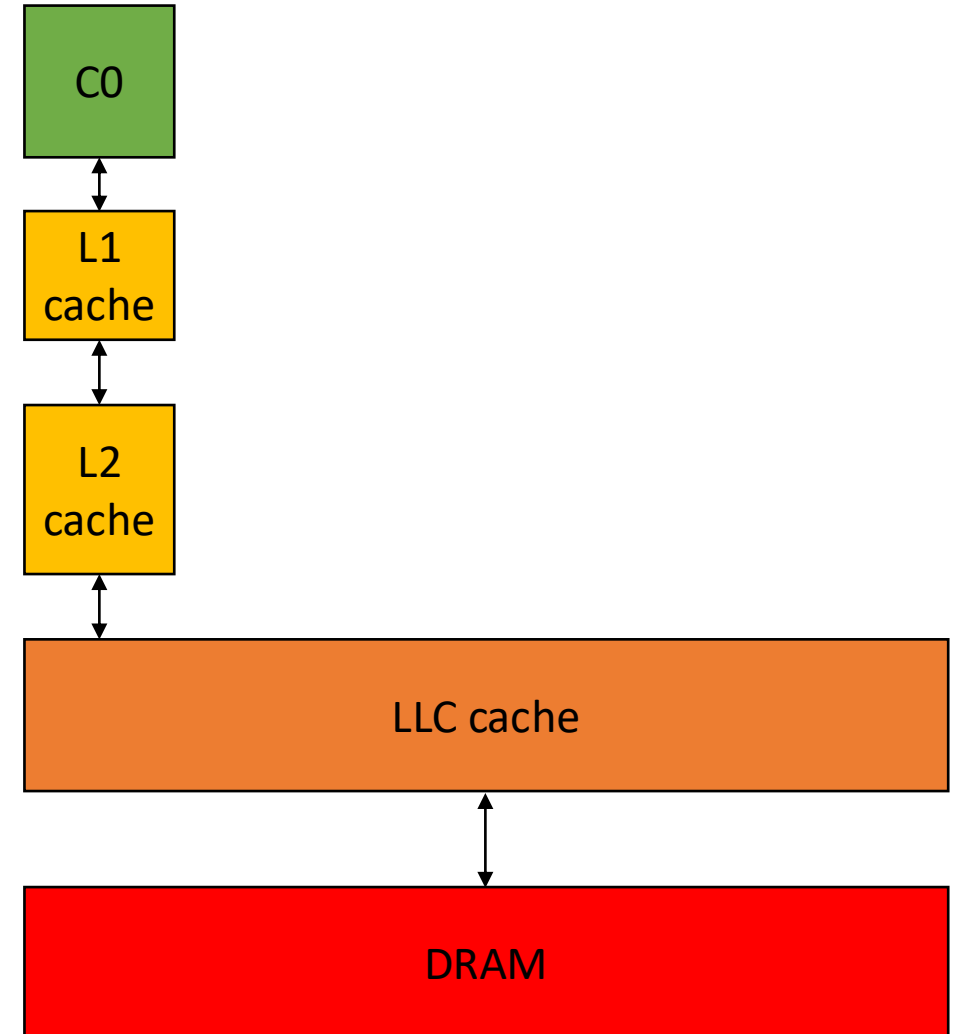


Caches

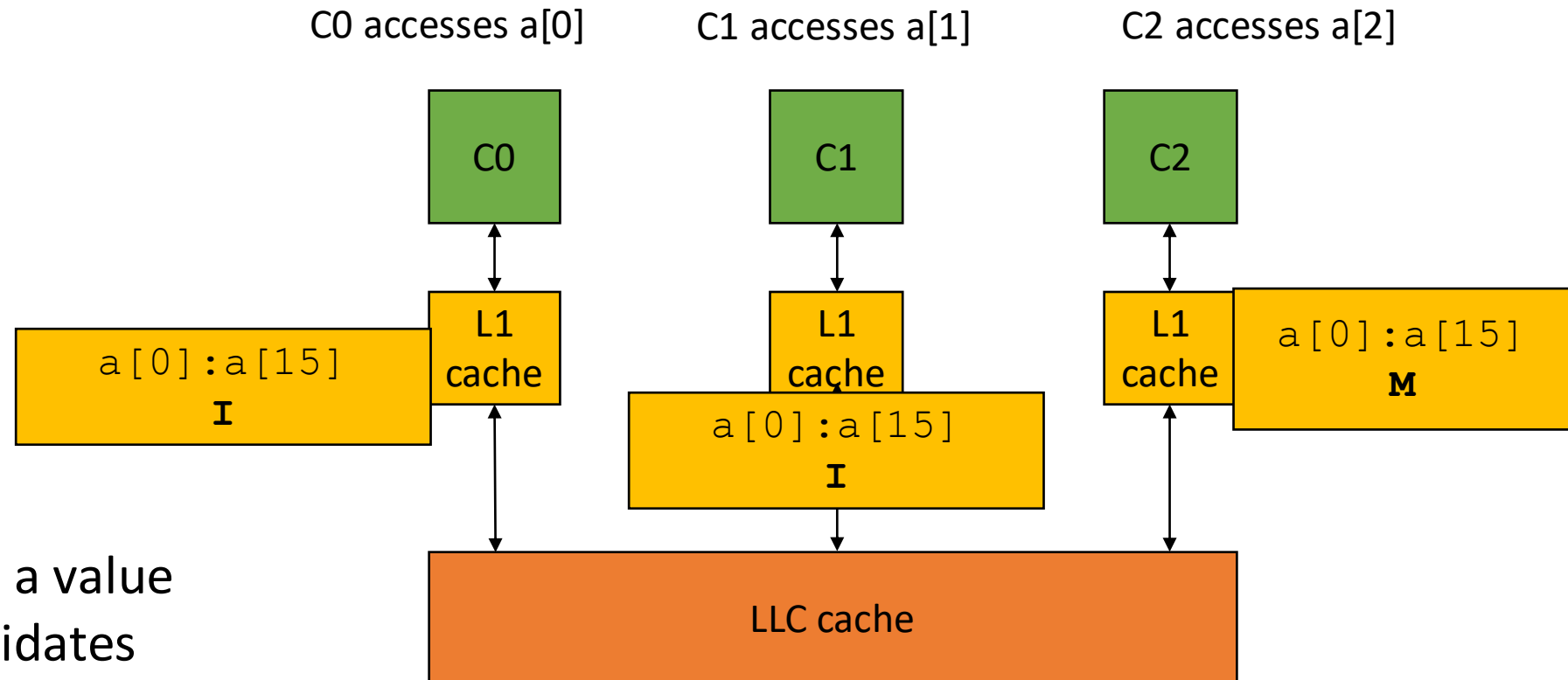
```
int increment(int *a) {  
    a[0]++;  
}
```

%5 = load i32, i32* %4	4 cycles
%6 = add nsw i32 %5, 1	1 cycles
store i32 %6, i32* %4	4 cycles

9 cycles!



Cache Coherence and False Sharing



when one core modifies a value in the cache line, it invalidates everyone else's cache line.

This is called ***False Sharing***

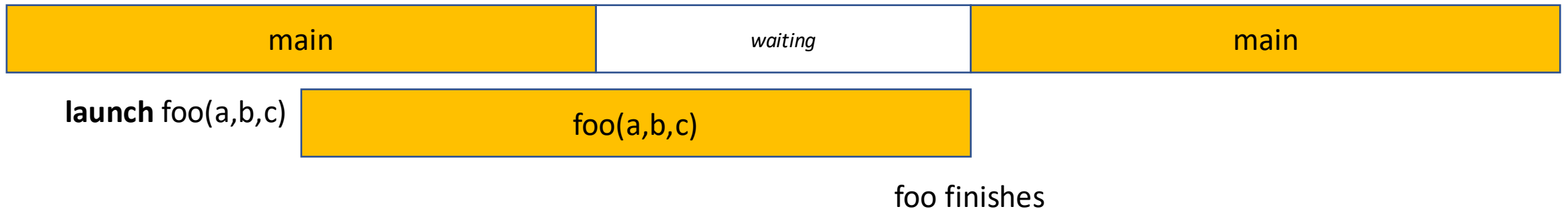
```
#include <thread>
using namespace std;

void foo(int a, int b, int c) {
    // some foo code
}

int main() {
    // some main code
    thread thread_handle (foo,1,2,3);
    // code here runs concurrently with foo
    thread_handle.join();
    return 0;
}
```

main waits for foo.
called **join()**

join() returns in main



```
#include <thread>
#include <iostream>
using namespace std;

void foo(int a, int b, int *c) {
    // return a + b;
    *c = a + b;
}

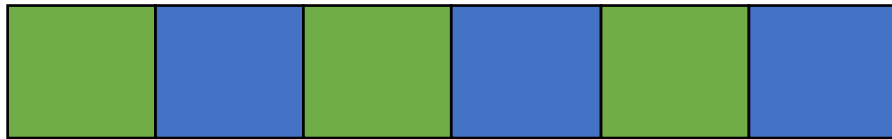
int main() {
    // some main code
    int ret = 0;
    thread thread_handle (foo, 1, 2, &ret);
    // code here runs concurrently with foo
    cout << ret << endl;
    thread_handle.join();
    return 0;
}
```

What if....

SPMD programming model

```
void increment_array(int *a, int a_size, int tid, int num_threads) {  
    for (int i = tid; i < a_size; i+=num_threads) {  
        a[i]++;  
    }  
}
```

iterations computed by thread 1



array a

switch to thread 1

Assume 2 threads
lets step through thread 1
i.e.
tid = 1
num_threads = 2

SPMD programming model

```
void increment_array(int *a, int a_size, int tid, int num_threads);
```

```
#define THREADS 8
#define A_SIZE 1024
int main() {
    int *a = new int[A_SIZE];
    // initialize a
    thread thread_ar[THREADS];

    for (int i = 0; i < THREADS; i++)
        thread_ar[i] = thread(increment_array, a, A_SIZE, i, THREADS);
    for (int i = 0; i < THREADS; i++)
        thread_ar[i].join();

    delete[] a;
    return 0;
}
```

New material

Embarrassingly parallel

Embarrassingly parallel

Embarrassingly parallel

From Wikipedia, the free encyclopedia

In [parallel computing](#), an **embarrassingly parallel** workload or problem (also called **embarrassingly parallelizable**, **perfectly parallel**, **delightfully parallel** or **pleasingly parallel**) is one where little or no effort is needed to separate the problem into a number of parallel tasks.^[1] This is often the case where there is little or no dependency or need for communication between those parallel tasks, or for results between them.^[2]

For this class: A multithreaded program is ***embarrassingly parallel*** if there are no ***data-conflicts***.

A ***data conflict*** is where one thread writes to a memory location that another thread reads or writes to concurrently and without sufficient ***synchronization***.

Embarrassingly parallel

- Consider the following program:

There are 3 arrays: a , b , c .

We want to compute $c[i] = a[i] + b[i]$

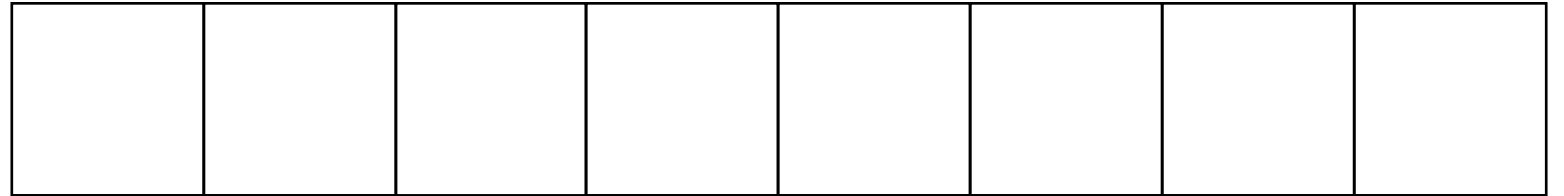
Embarrassingly parallel

array a



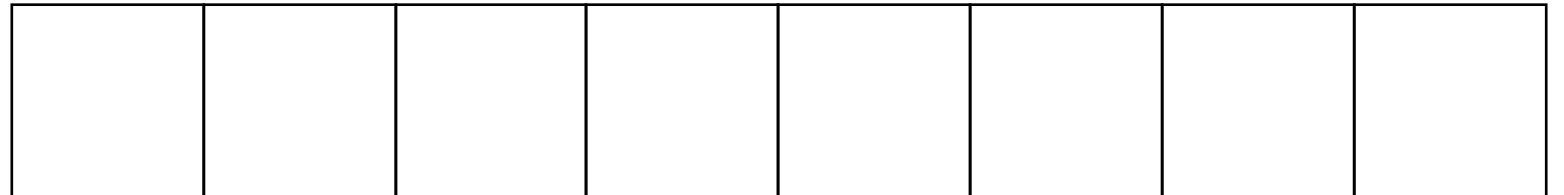
+ + + + + + + +

array b



= = = = = = = =

array c



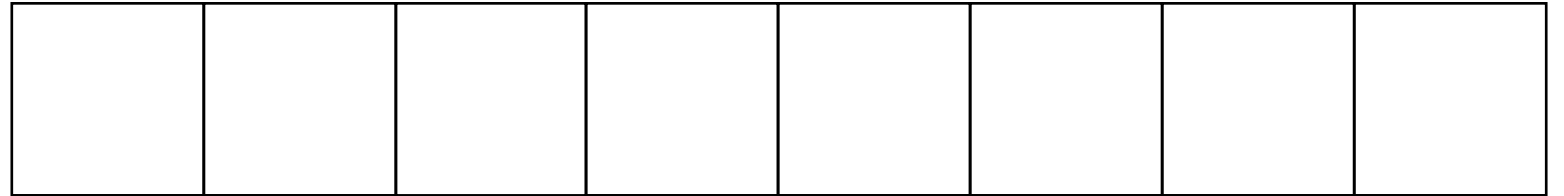
Embarrassingly parallel

array a



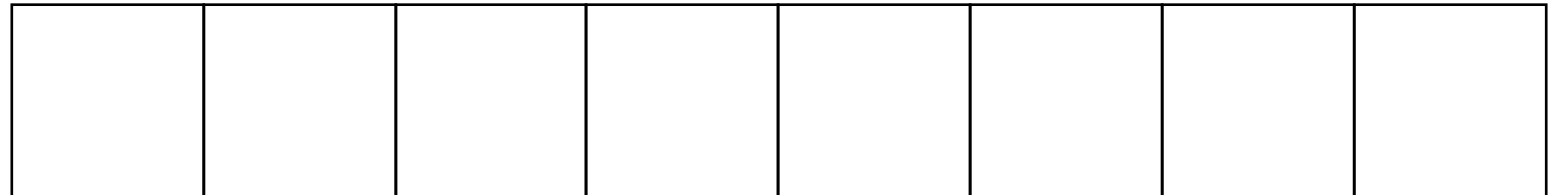
+ + + + + + + +

array b



= = = = = = = =

array c

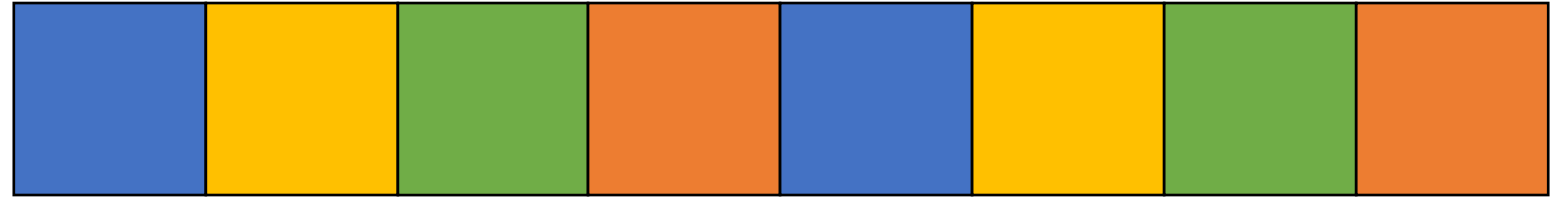


Computation
can easily be
divided into
threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

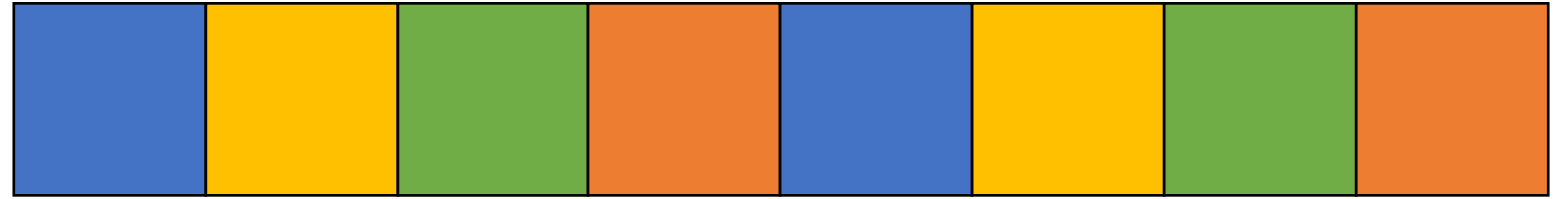
Embarrassingly parallel

array a



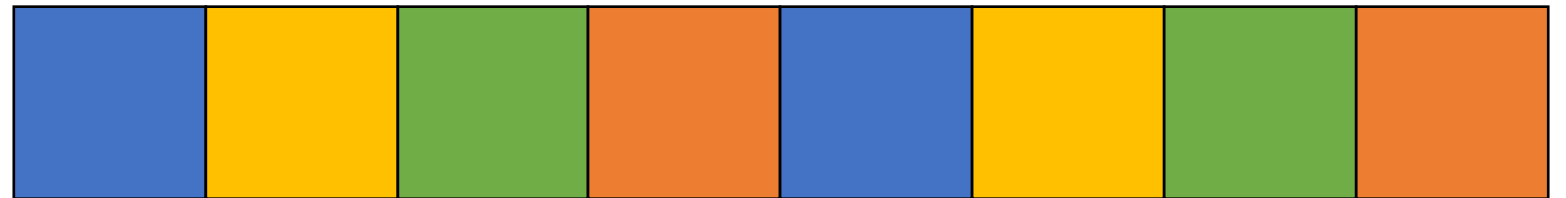
+ + + + + + + +

array b



= = = = = = = =

array c



Computation
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threads

Thread 0 - Blue
Thread 1 - Yellow
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Thread 3 - Orange

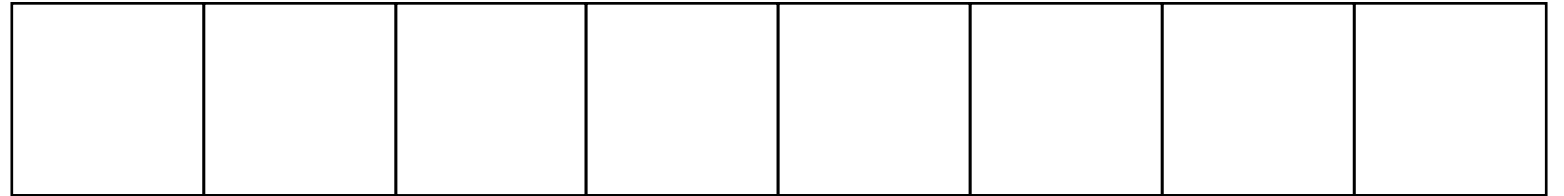
Embarrassingly parallel

array a



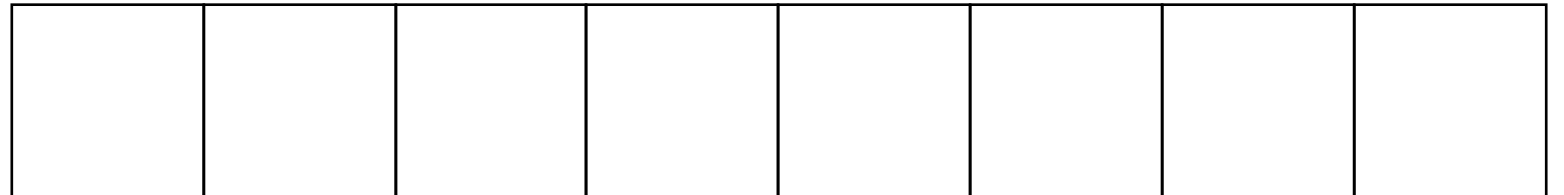
+ + + + + + + +

array b



= = = = = = = =

array c



Computation
can easily be
divided into
threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

Embarrassingly parallel

array a



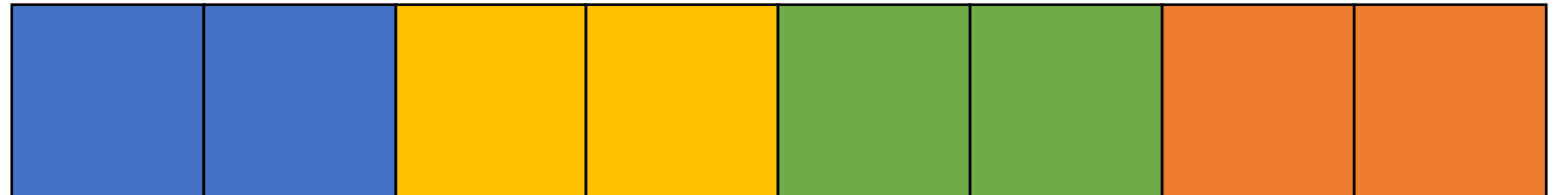
+ + + + + + + +

array b



= = = = = = = =

array c



Computation
can easily be
divided into
threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

Embarrassingly parallel

- The different parallelization strategies will probably have different performance behaviors.
- But they are both embarrassingly parallel solutions to the problem
- There is lots of research into making these types of programs go fast!
 - but this module will focus on programs that require synchronization

Embarrassingly parallel

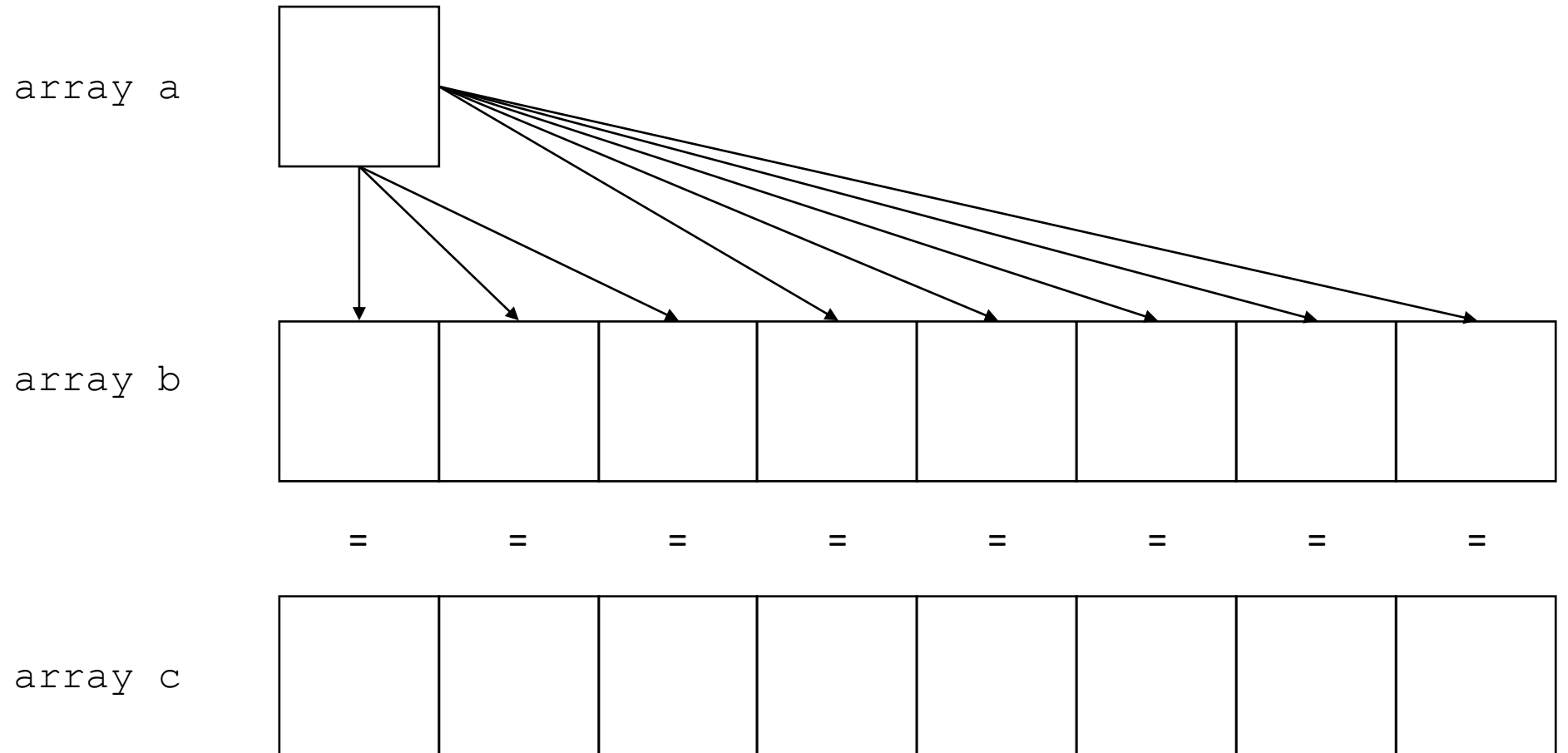
- Next Program

There are 3 arrays: a , b , c .

We want to compute $c[i] = a[0] + b[i]$

Embarrassingly parallel

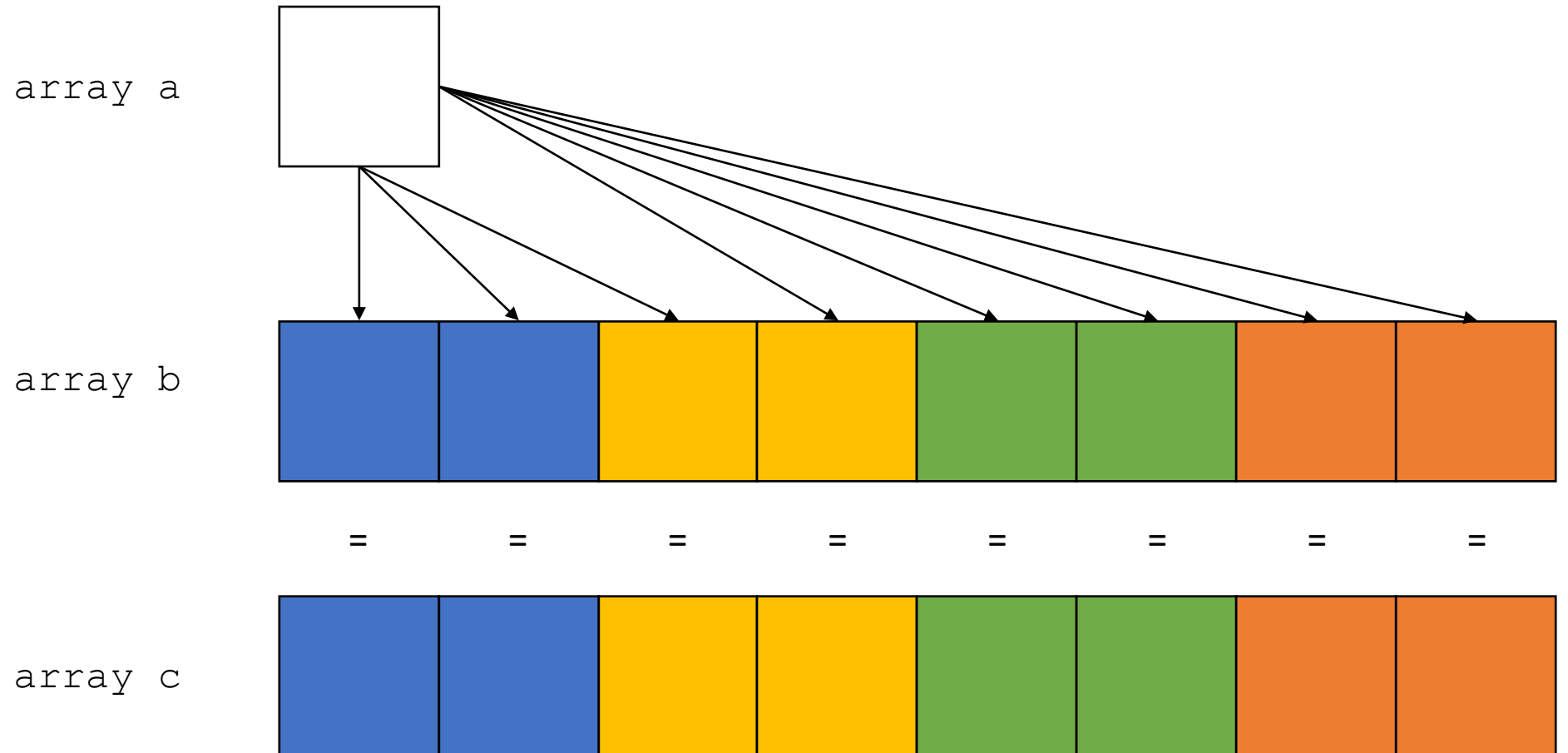
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange



*is this problem
embarrassingly
parallel?*

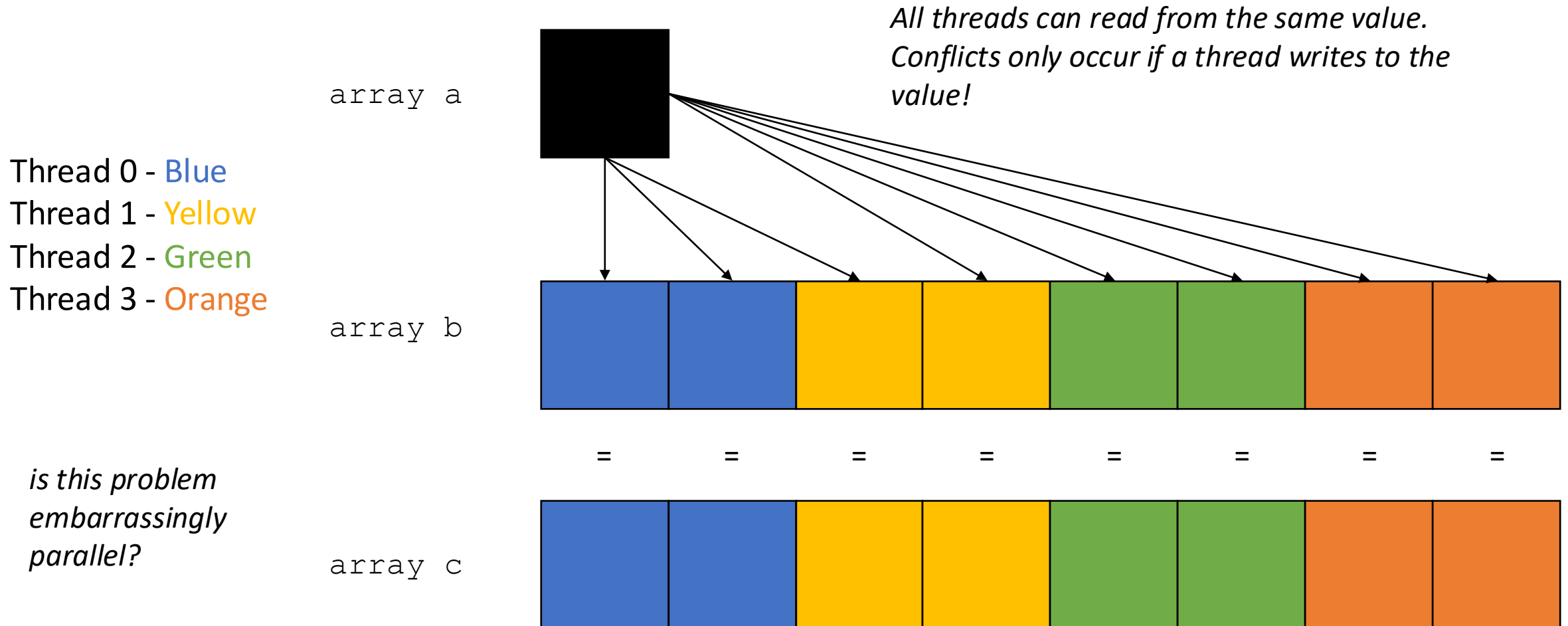
Embarrassingly parallel

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange



*is this problem
embarrassingly
parallel?*

Embarrassingly parallel



Embarrassingly parallel

- Next Program

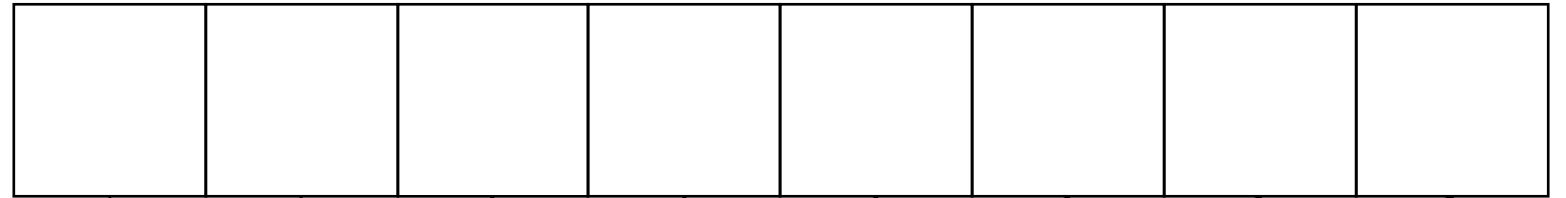
There are 2 arrays: b , c

We want to compute $c[0] = b[0] + b[1] + b[2] \dots$

Embarrassingly parallel

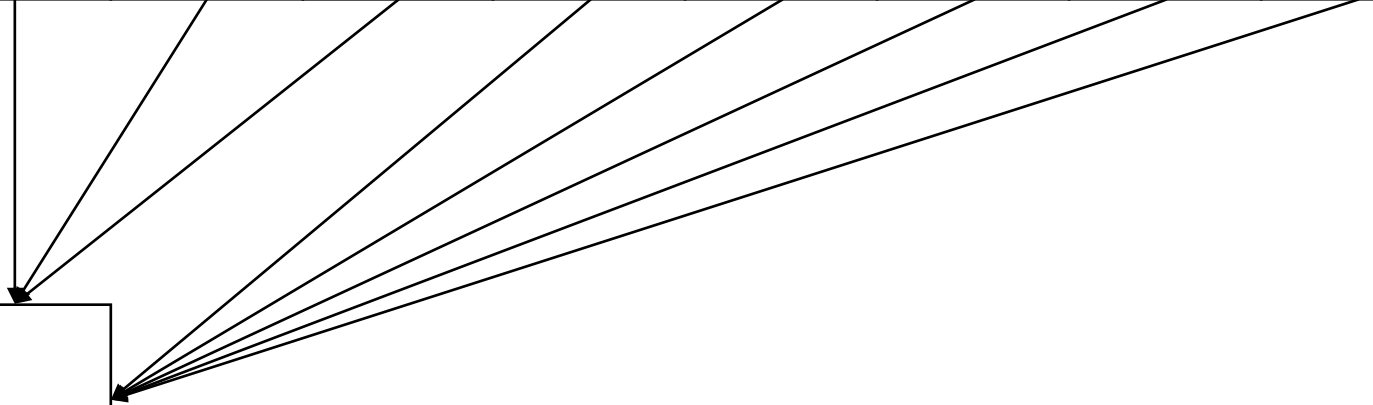
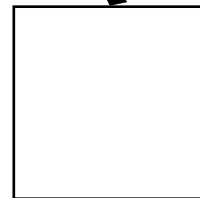
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

array b



*is this problem
embarrassingly
parallel?*

array c



Embarrassingly parallel

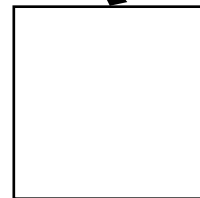
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

array b



*is this problem
embarrassingly
parallel?*

array c



*threads read
unique locations*

Embarrassingly parallel

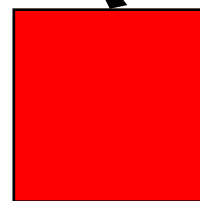
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

array b



*is this problem
embarrassingly
parallel?*

array c



*threads read
unique locations*

Conflict because multiple threads write to the same location!

Embarrassingly parallel

Note: Reductions have some parallelism in them, as seen in your homework.

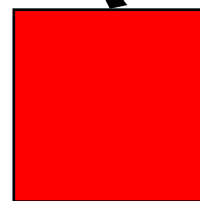
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

array b



*is this problem
embarrassingly
parallel?*

array c



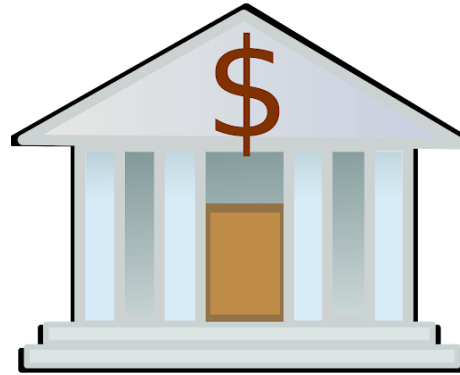
*threads read
unique locations*

Conflict because multiple threads write to the same location!

We need a way to safely share memory

We need a way to safely share memory

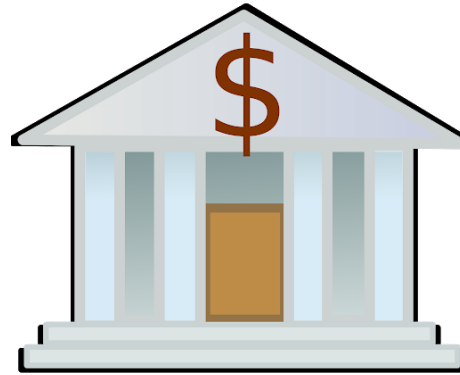
- Bank



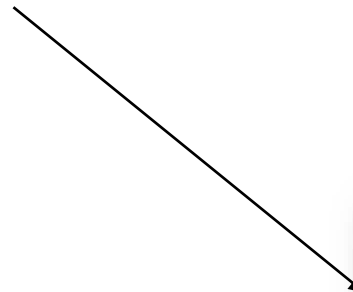
My account: \$\$

We need a way to safely share memory

- Bank

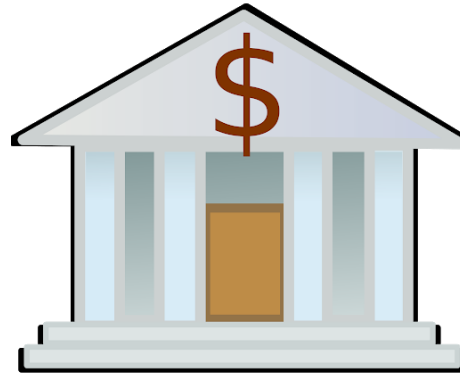


My account: \$\$



We need a way to safely share memory

- Bank



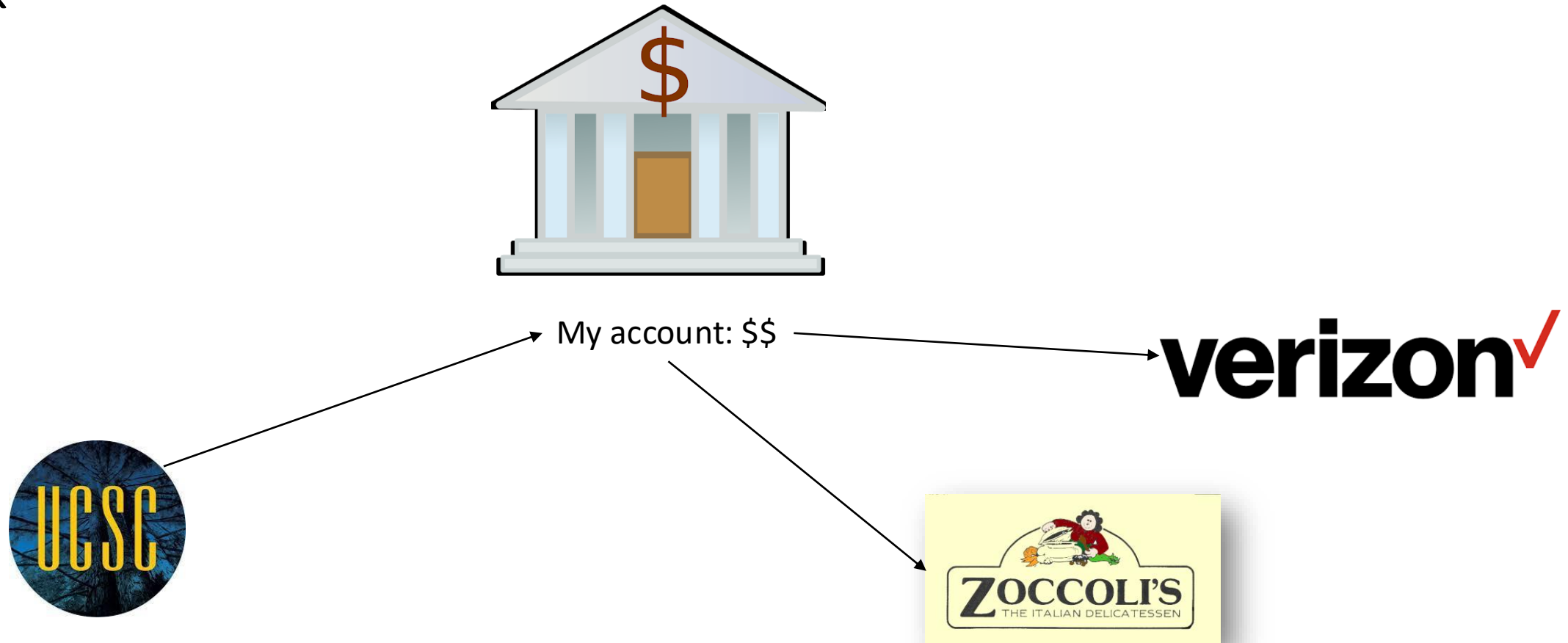
My account: \$\$

verizon✓



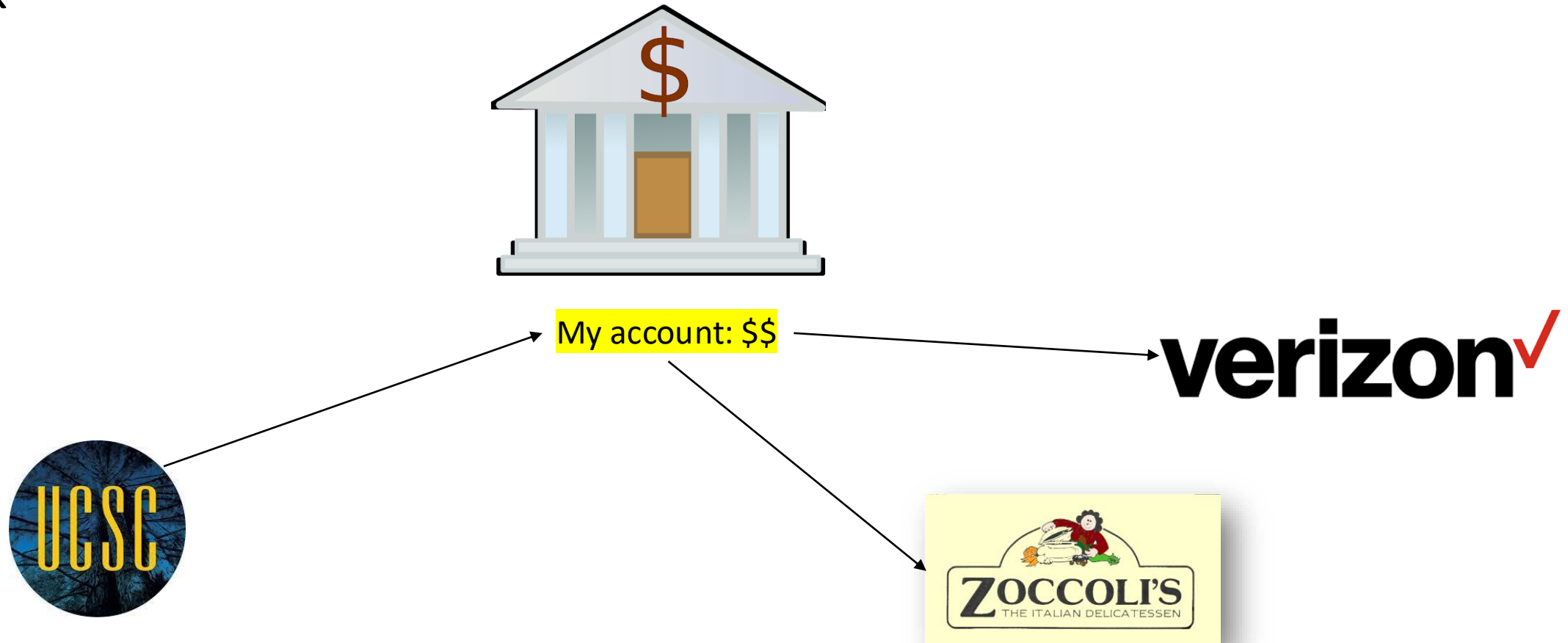
We need a way to safely share memory

- Bank



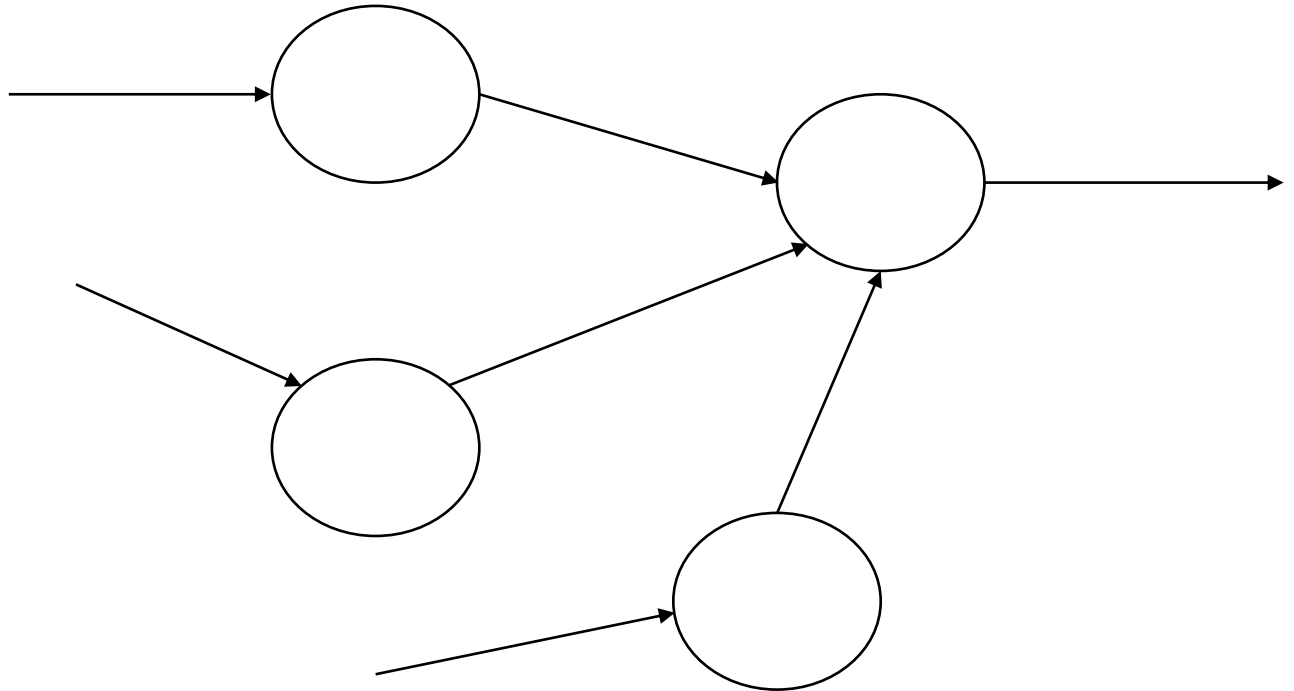
We need a way to safely share memory

- Bank



We need a way to safely share memory

- Graph algorithms

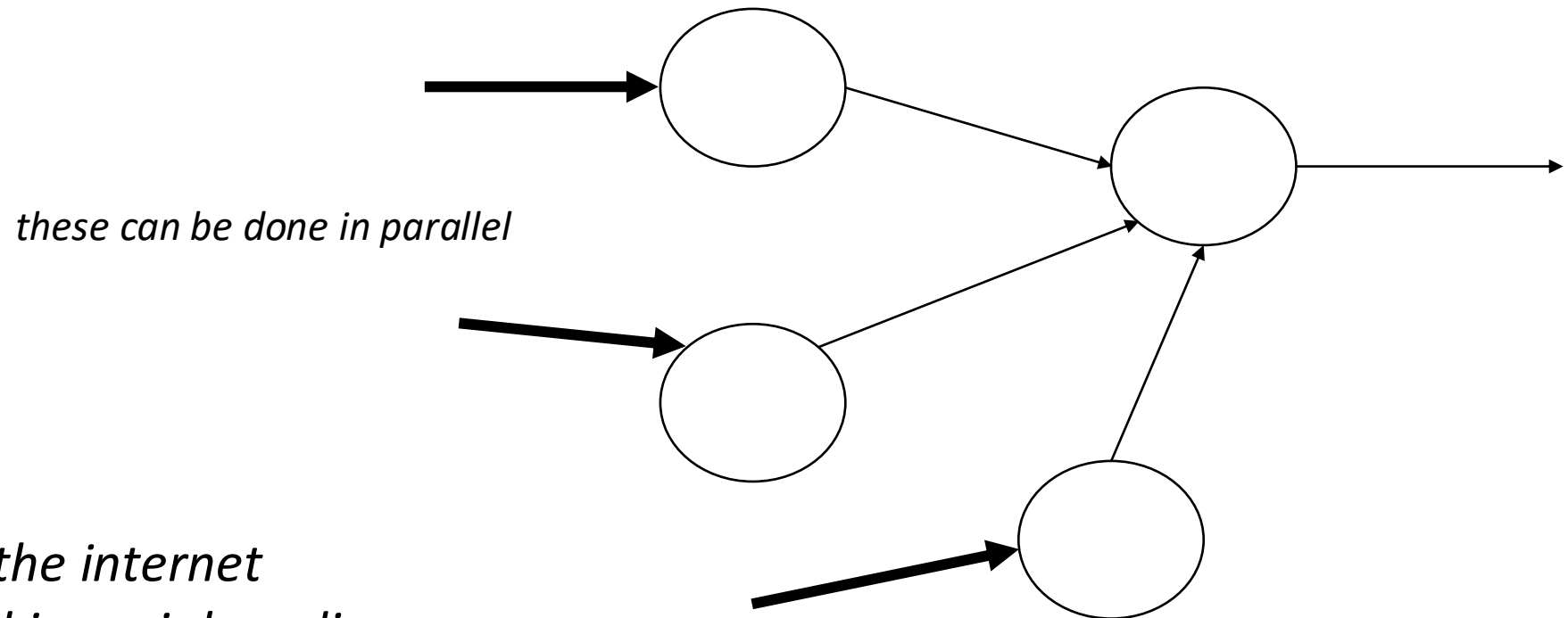


Examples:

*Ranking pages on the internet
information spread in social media*

We need a way to safely share memory

- Graph algorithms



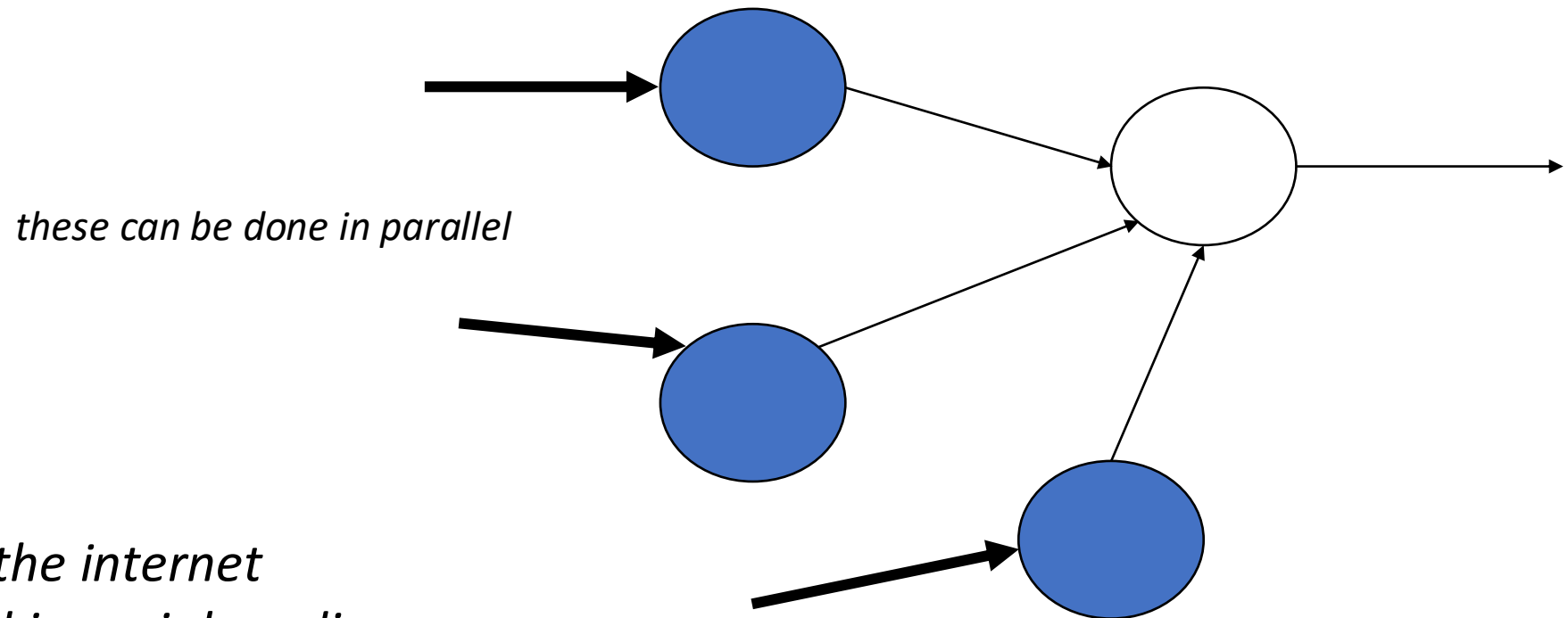
Examples:

Ranking pages on the internet

information spread in social media

We need a way to safely share memory

- Graph algorithms

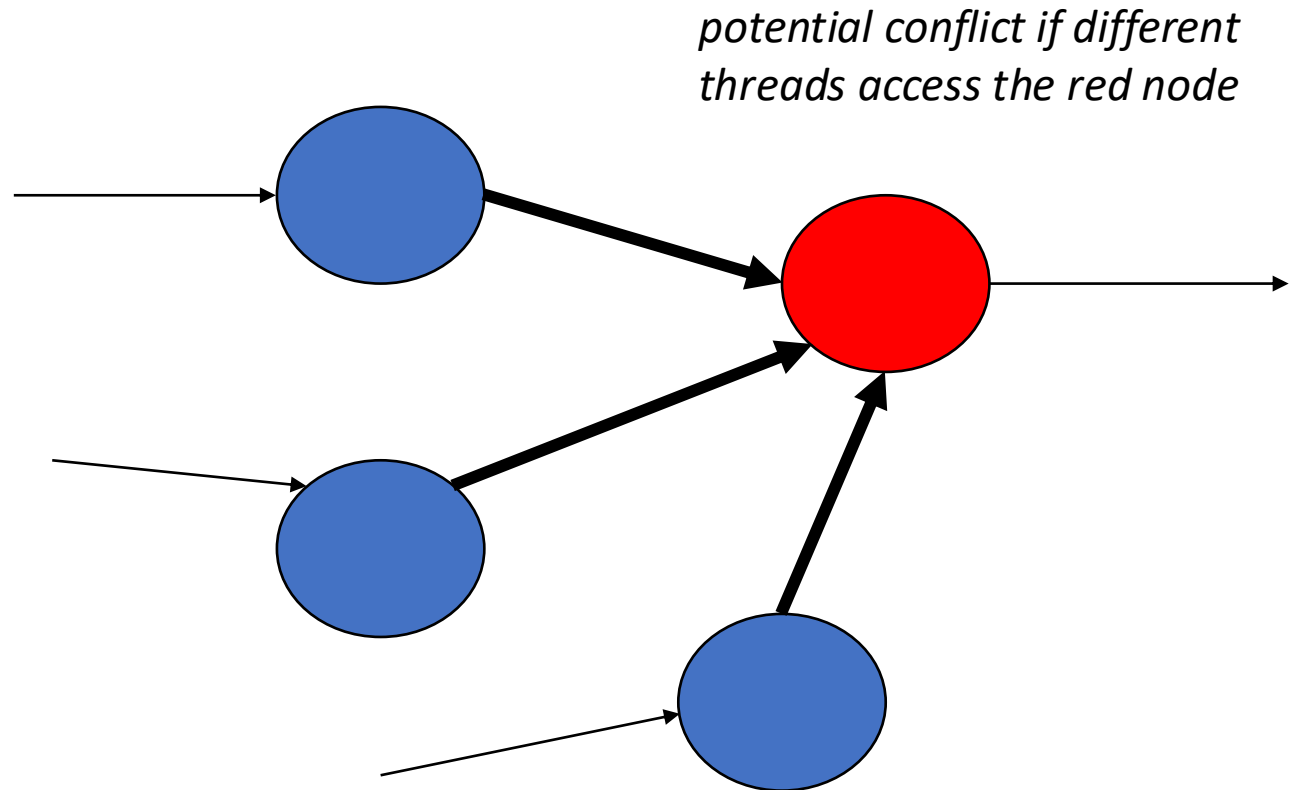


Examples:

*Ranking pages on the internet
information spread in social media*

We need a way to safely share memory

- Graph algorithms



Examples:

*Ranking pages on the internet
information spread in social media*

We need a way to safely share memory

- Machine Learning

The diagram shows a dot product calculation. A yellow curved arrow labeled "Dot Product" connects the first row of the first matrix to the first column of the second matrix. The first matrix is $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and the second matrix is $\begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$. The result is a single value, 58, shown in a yellow circle inside a blue bracket.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 \end{bmatrix}$$

Lots of machine learning is some form of matrix multiplication

We need a way to safely share memory

- Machine Learning

conflict!

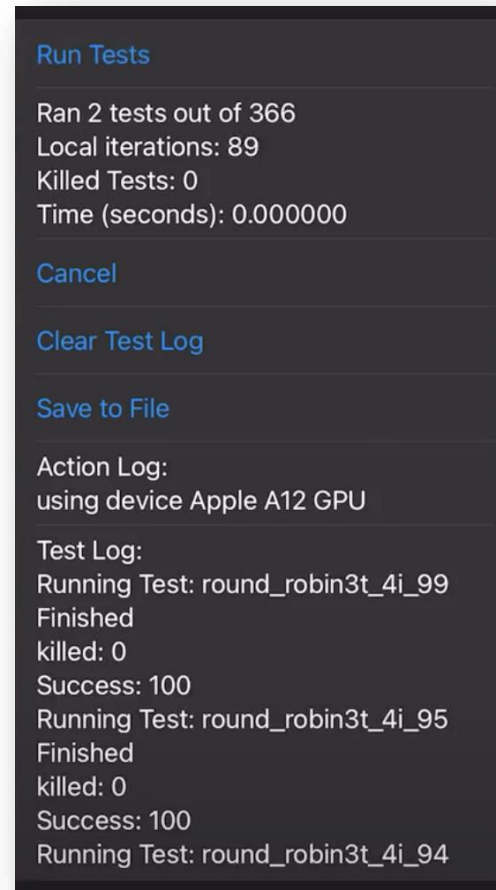
"Dot Product"

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 \\ \dots \end{bmatrix}$$

Lots of machine learning is some form of matrix multiplication

We need a way to safely share **resources**

- User interfaces



*background process
that provides progress
updates to the UI.*

*UI updates must be
synchronized!!*

<https://drive.google.com/file/d/1JVQTTQsrKhpksgVAM1yaMQkyohfDtWsSI/view?usp=sharing>

Dangers of conflicts

- We will illustrate using a running bank account example

Sequential bank scenario

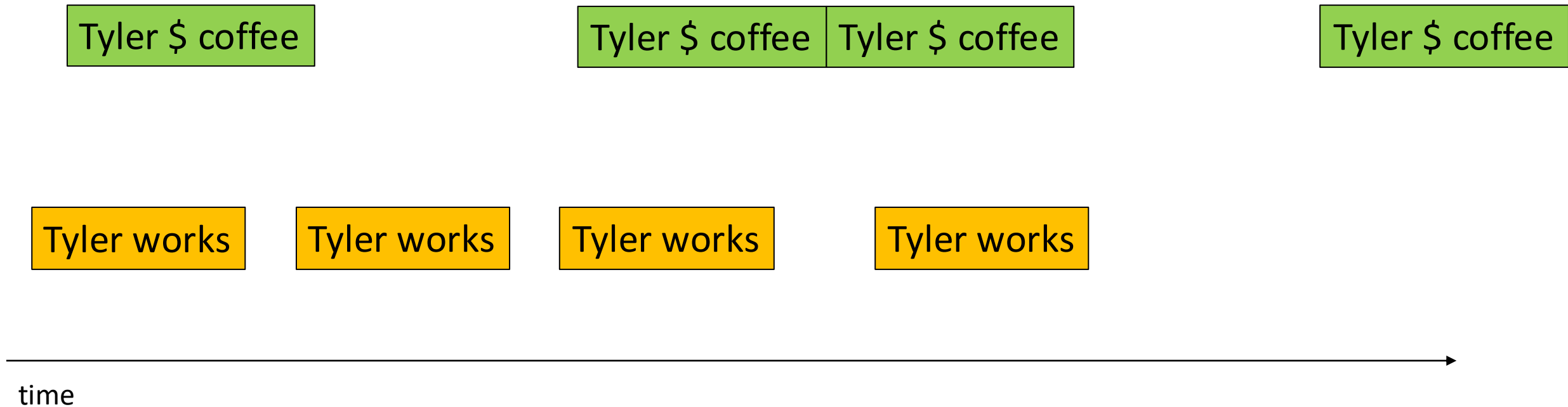
- UCSC deposits \$1 in my bank account after every hour I work.
- I buy a cup of coffee (\$1) after each hour I work.
- I work 1M hours (which is actually true).
- *I should break even*
- C++ code

Concurrent bank scenario

- UCSC contracts me to work 1M hours.
- My bank is so impressed with my contract that they give me a credit card. i.e. I can overdraw as long as I pay it back.
- UCSC deposits \$1 in my bank account **at some point** for every hour I work.
- I budget \$1M to spend on coffee **at some point** during work.

Concurrent bank scenario

This sets up a scheme where I buy coffee concurrently with working



Code demo

Reasoning about concurrency

- What is going on?
- We need to be able to reason more rigorously about concurrent programs

A thread is a sequential program

Tyler's coffee addiction:

```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```

Tyler's employer

```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```

A thread is a sequential program

Tyler's coffee addiction:

```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```

Tyler's employer

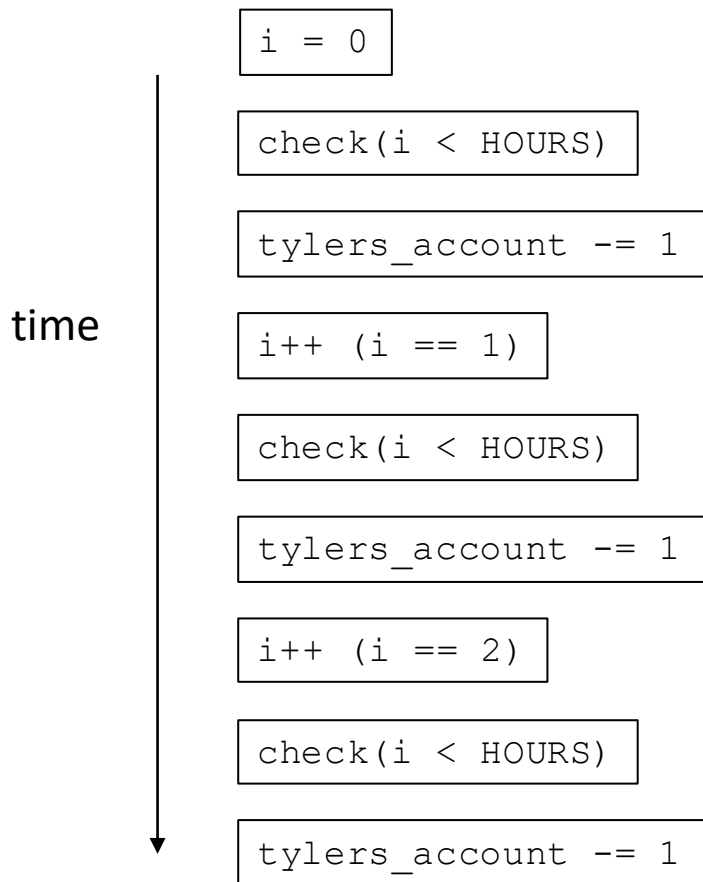
```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```

The execution of a program gives rise to events
Important distinction between program and events

A thread is a sequential program

Tyler's coffee addiction:

```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```



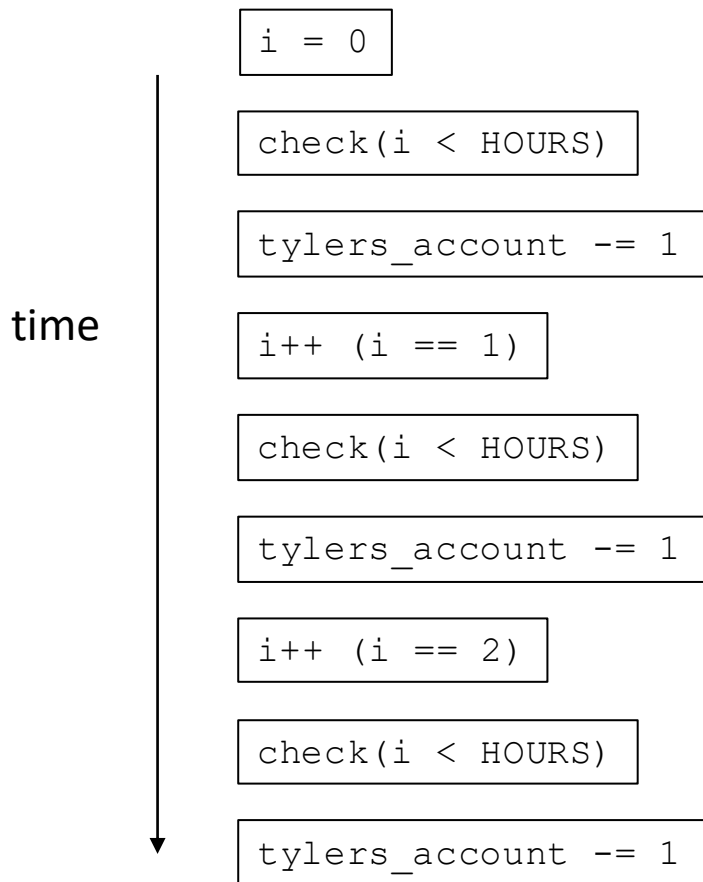
Tyler's employer

```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```

A thread is a sequential program

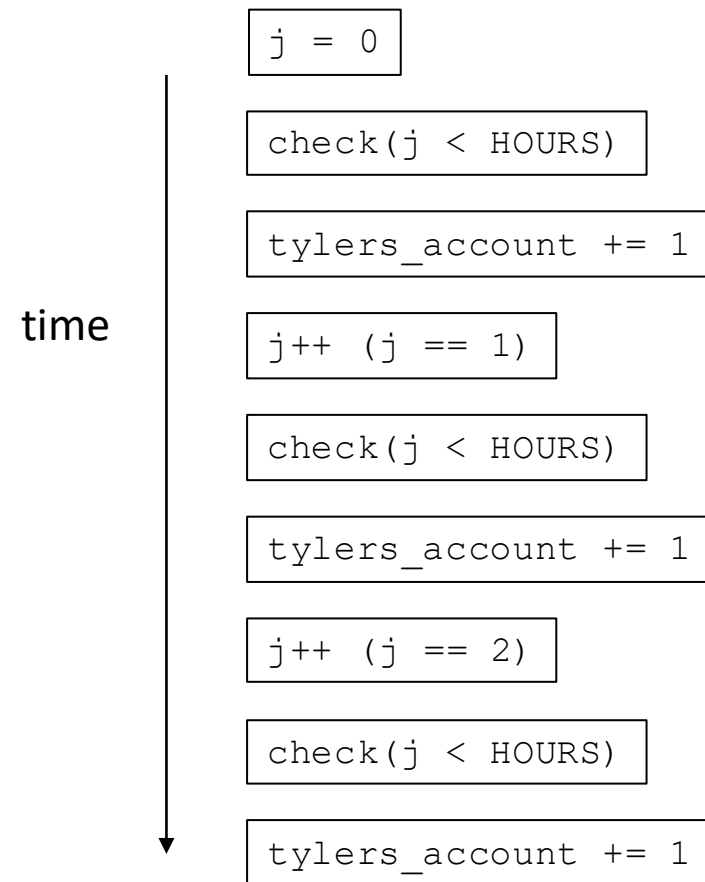
Tyler's coffee addiction:

```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```



Tyler's employer

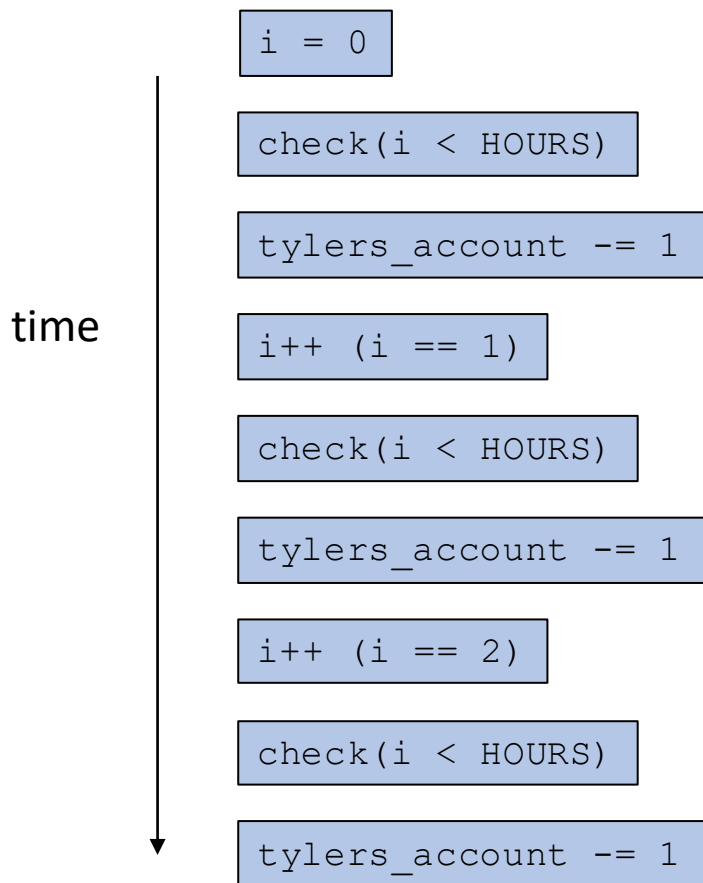
```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```



A thread is a sequential program

Tyler's coffee addiction:

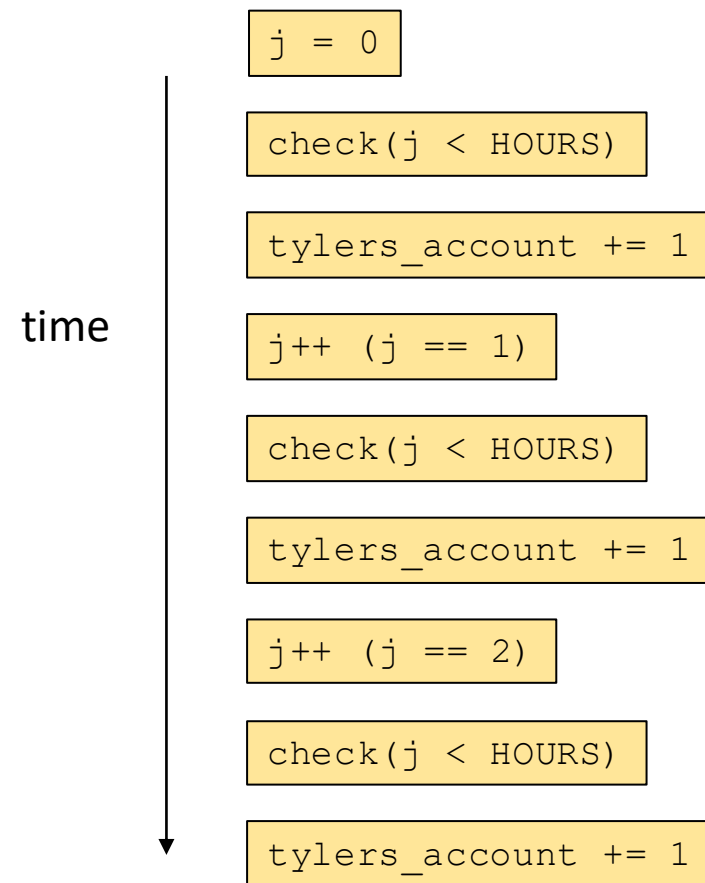
```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```



*color code events.
coffee thread is blue
payment thread is yellow*

Tyler's employer

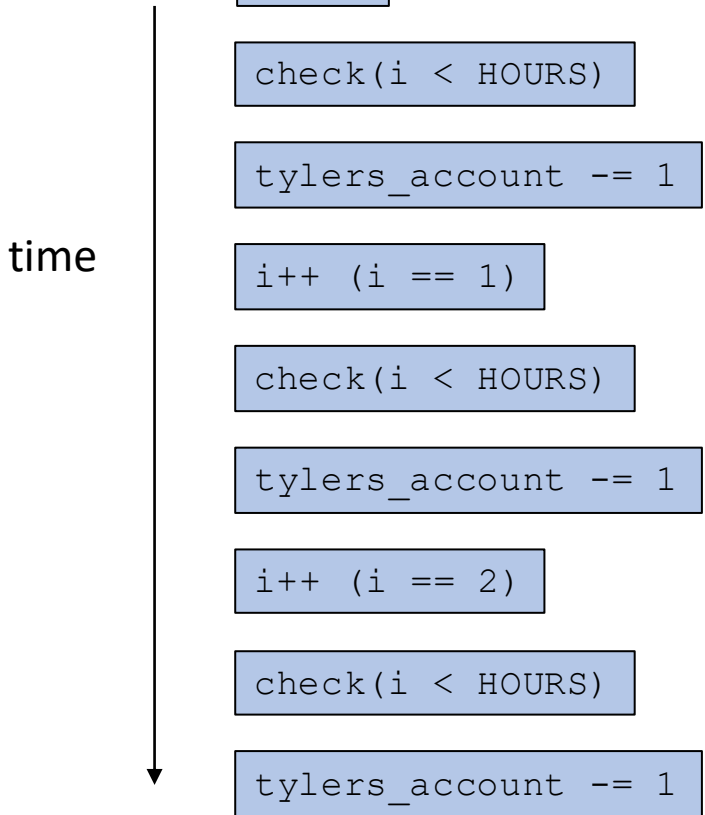
```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```



A thread is a sequential program

Tyler's coffee addiction:

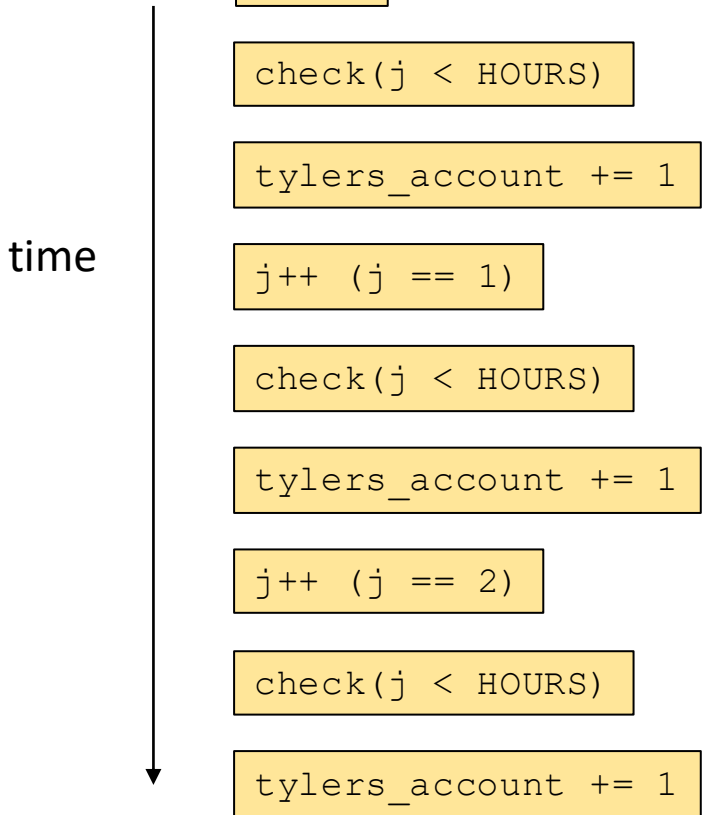
```
for (int i = 0; i < HOURS; i++) {  
    tylers_account -= 1;  
}
```

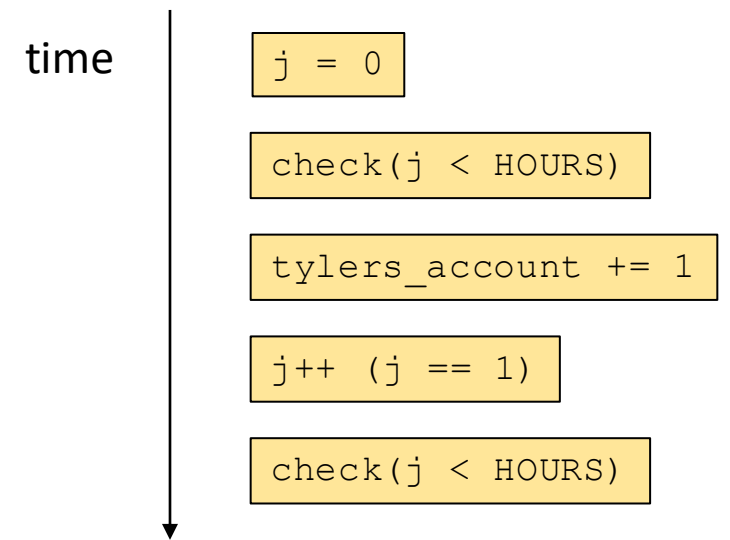
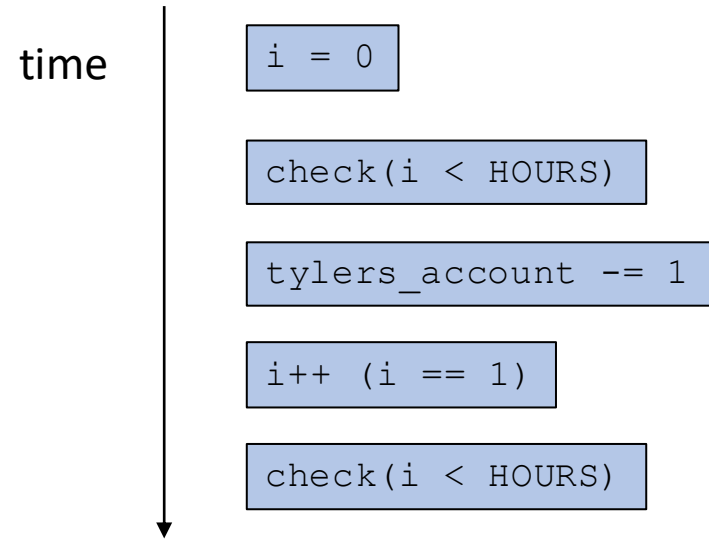


Any interleaving of the events is a valid execution of the concurrent program!

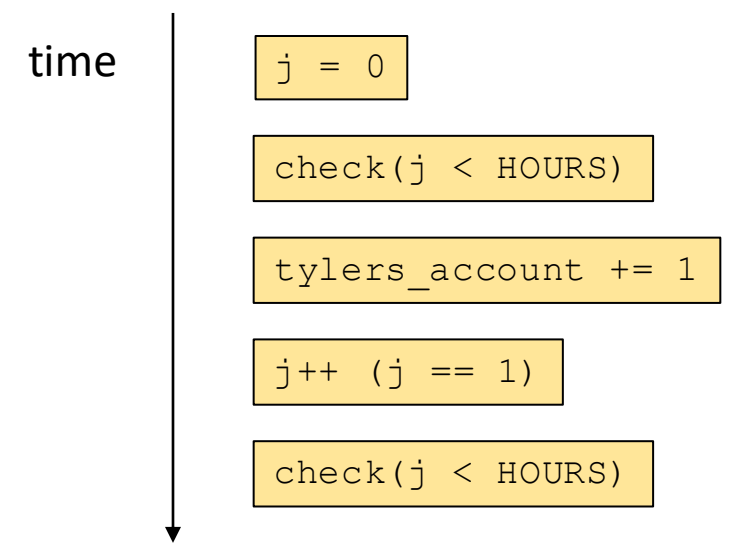
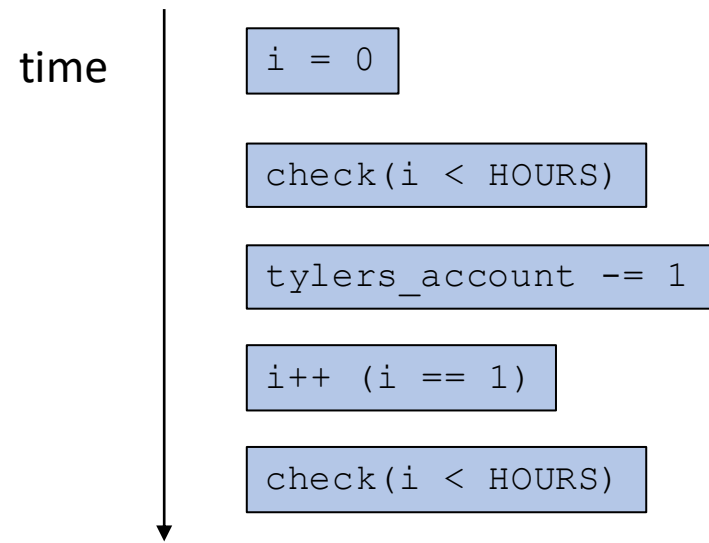
Tyler's employer

```
for (int j = 0; j < HOURS; j++) {  
    tylers_account += 1;  
}
```





consider just one loop iteration

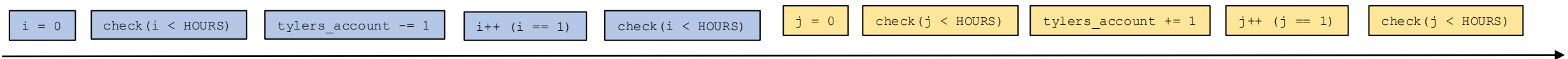


Concurrent execution



one possible execution

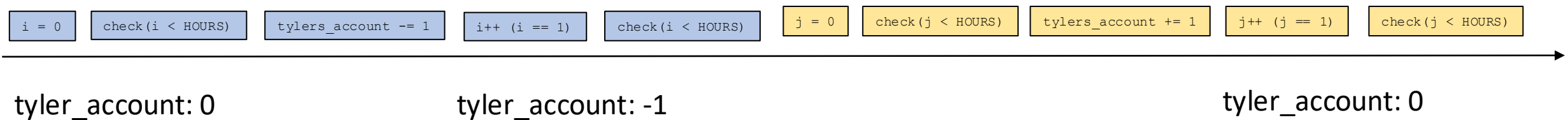
Concurrent execution

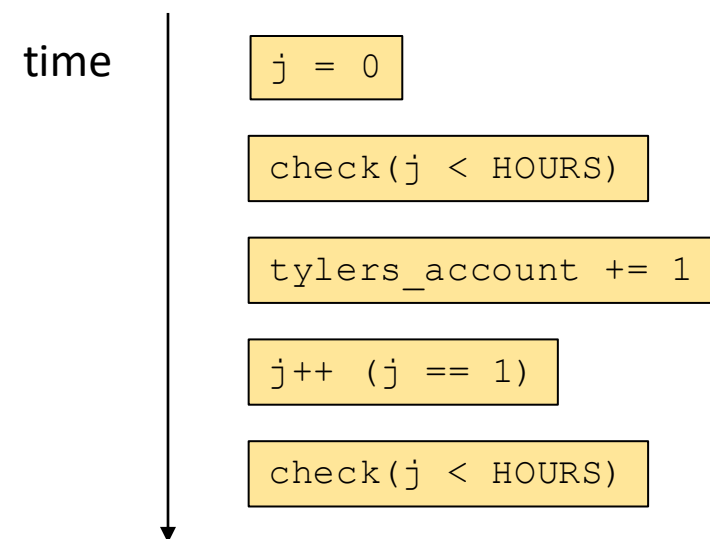
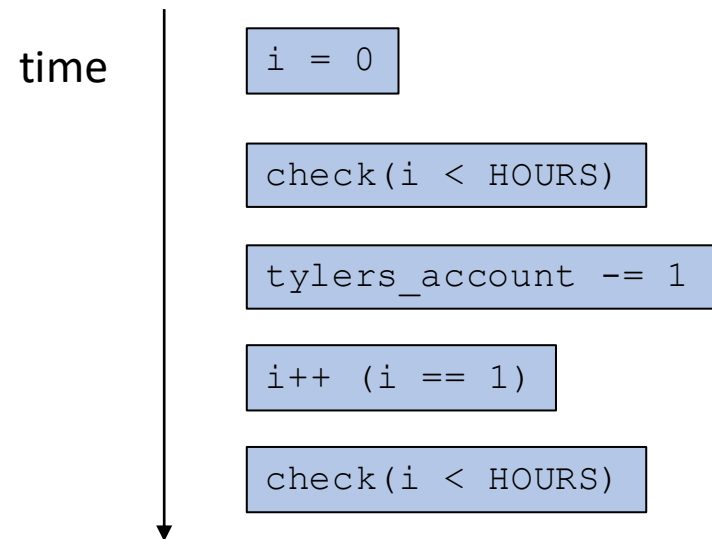




one possible execution

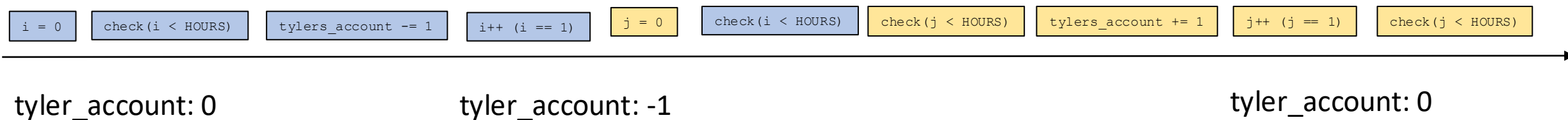
Concurrent execution

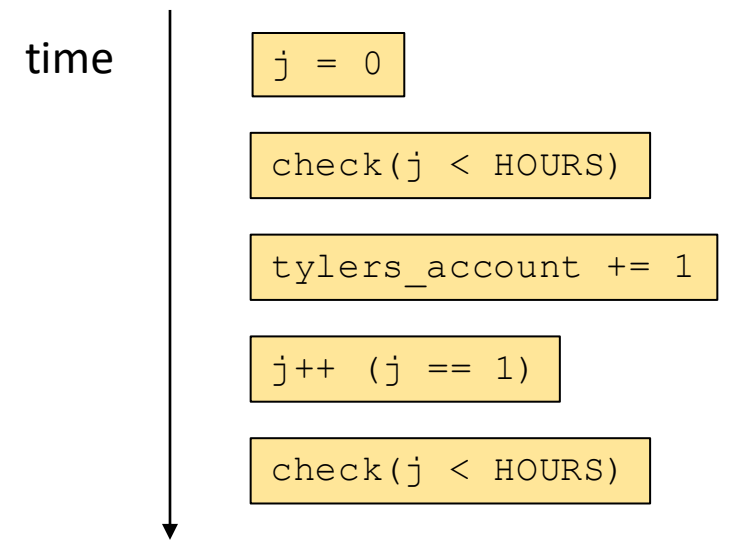
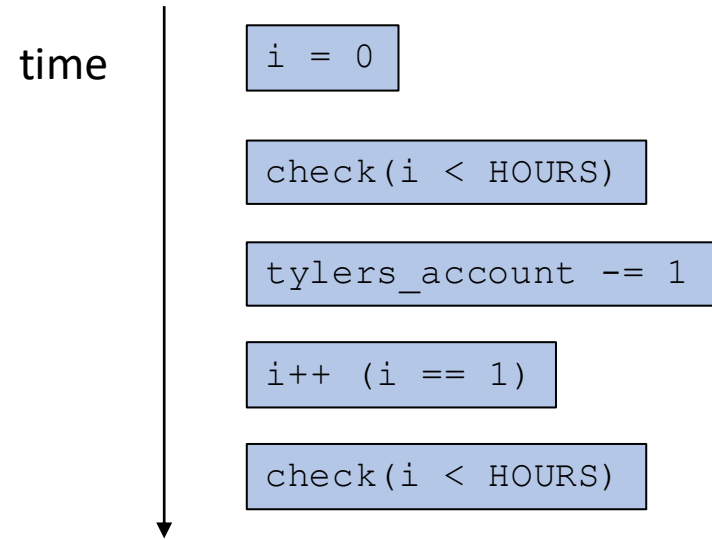




Another possible execution

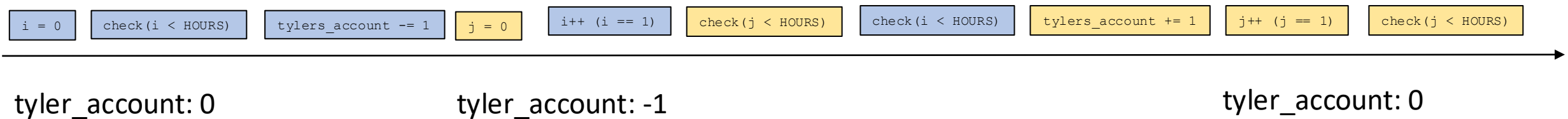
Concurrent execution

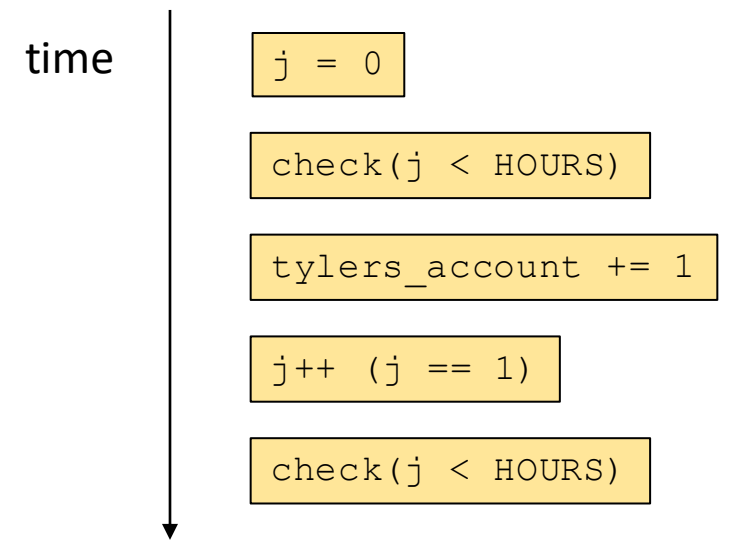
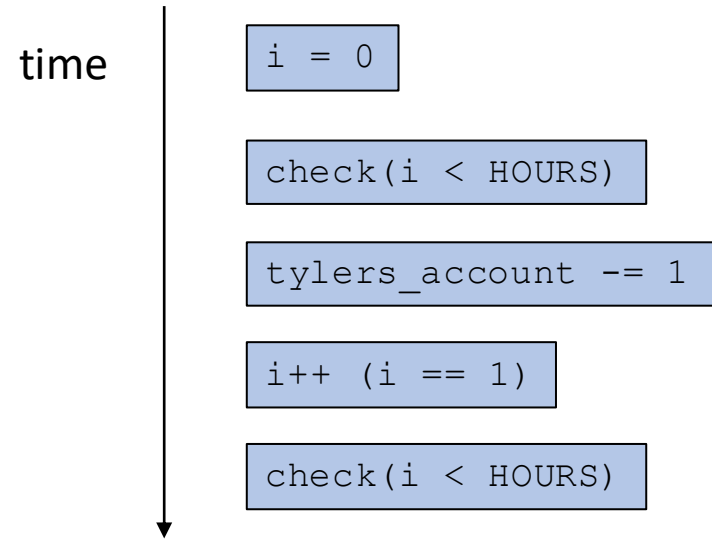




Another possible execution

Concurrent execution

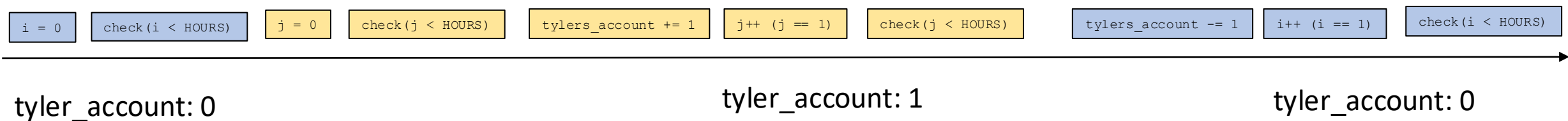


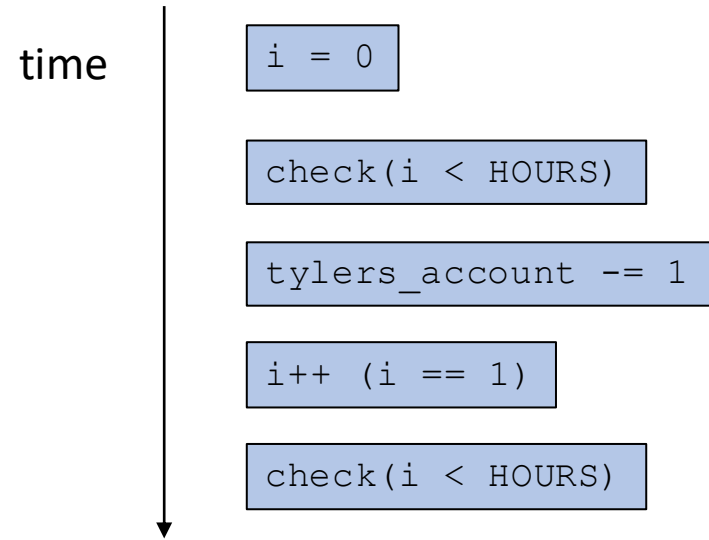


Another possible execution

Concurrent execution

This time my account isn't ever negative





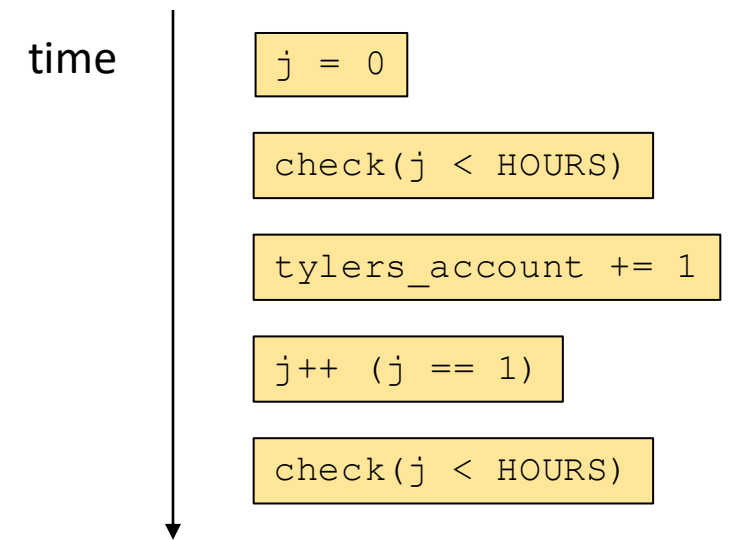
How many possible interleavings?

Combinatorics question:

if Thread 0 has N events

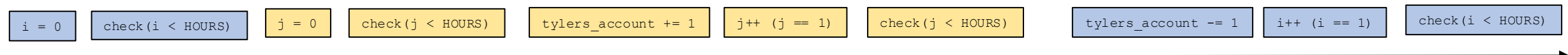
if Thread 1 has M events

$$\frac{(N + M)!}{N! M!}$$



Concurrent execution

in our example there are 252 possible interleavings!



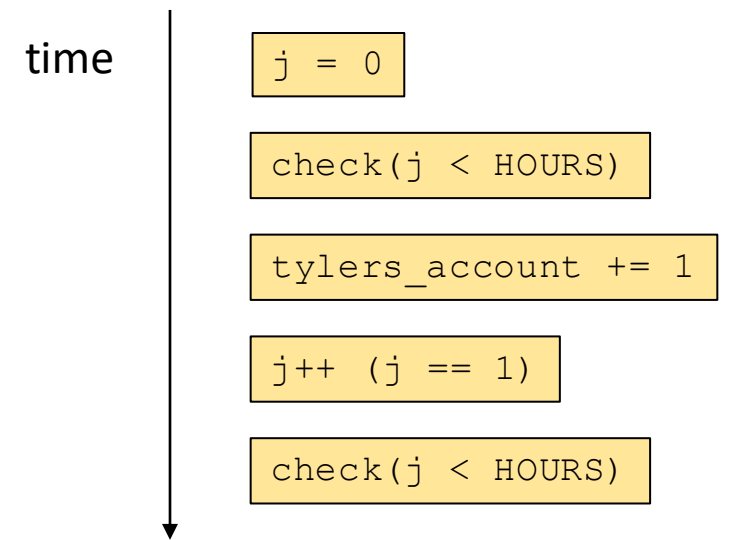
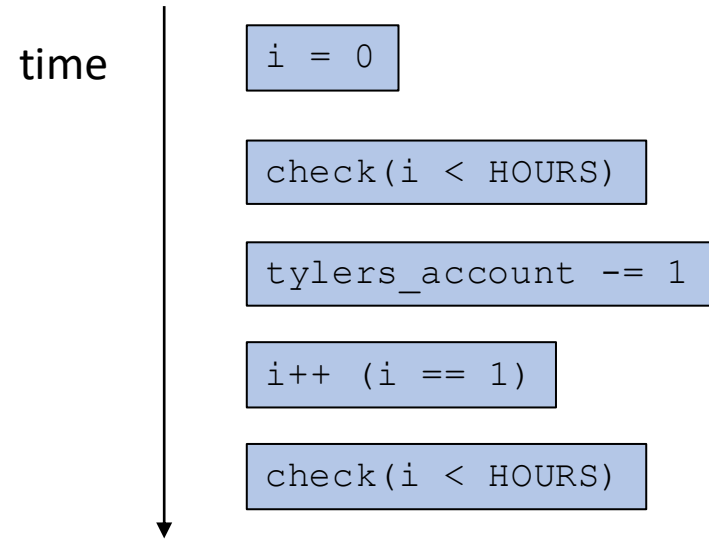
tyler_account: 0

tyler_account: 1

tyler_account: 0

Reasoning about concurrency

- Not feasible to think about all interleavings!
 - Lots of interesting research in pruning, testing interleavings
 - Very difficult to debug
- Think about smaller instances of the problem
- **Reduce the problem:** *If there's a problem, we should be able to see it in a single loop iteration.*

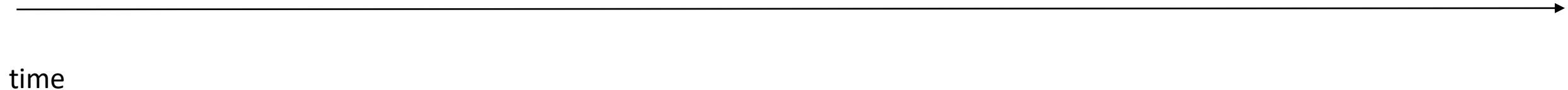


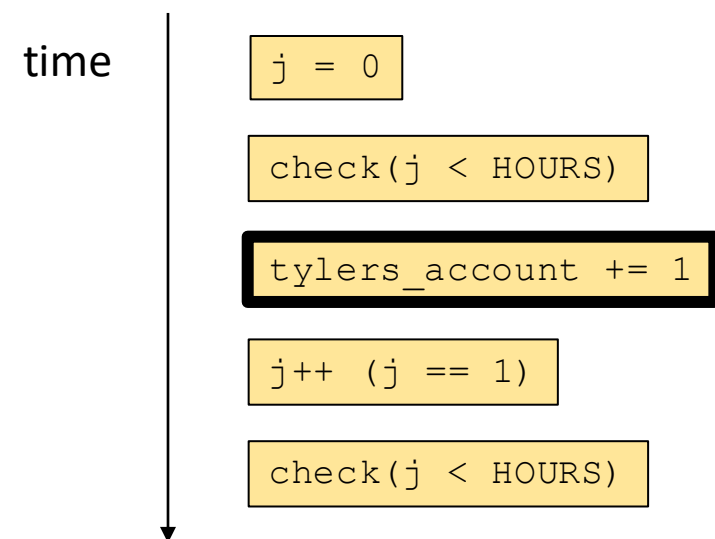
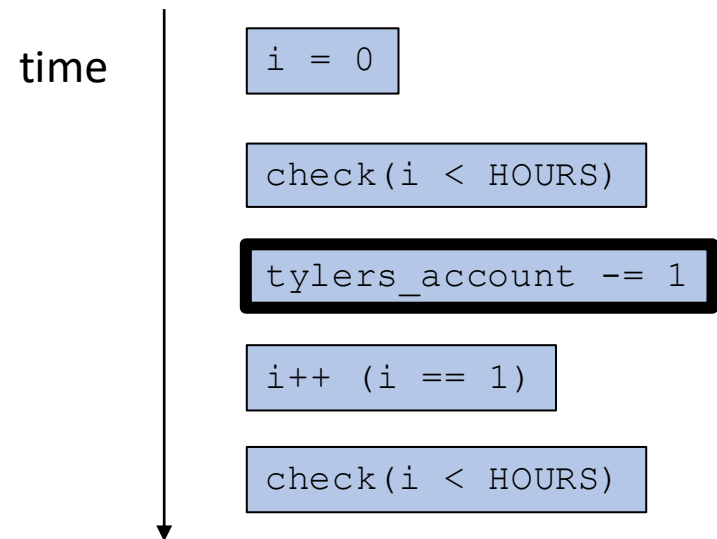
Lets get to the bottom of our money troubles:

For any interleaving, both of the increase and decrease must happen in some order.

So there isn't an interleaving that will explain the issue.

concurrent execution





concurrent execution



time

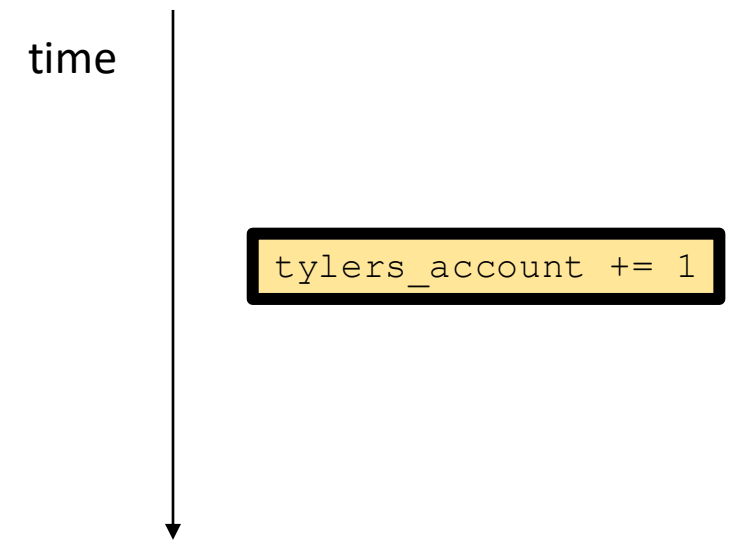
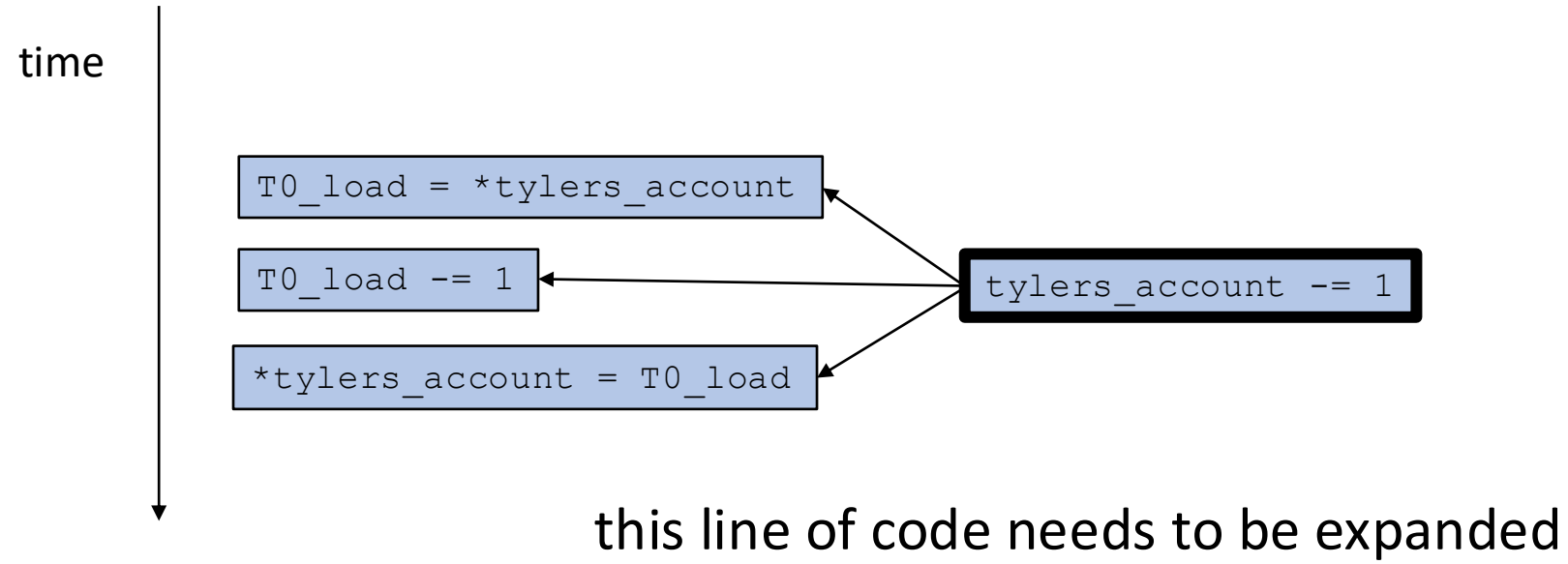


Remember 3 address code...

concurrent execution



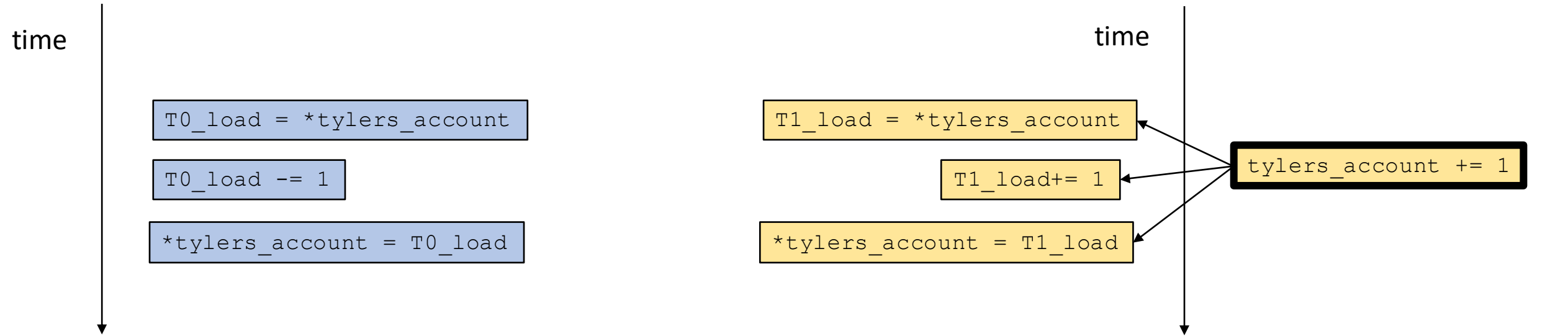
time



Remember 3 address code...

concurrent execution

time

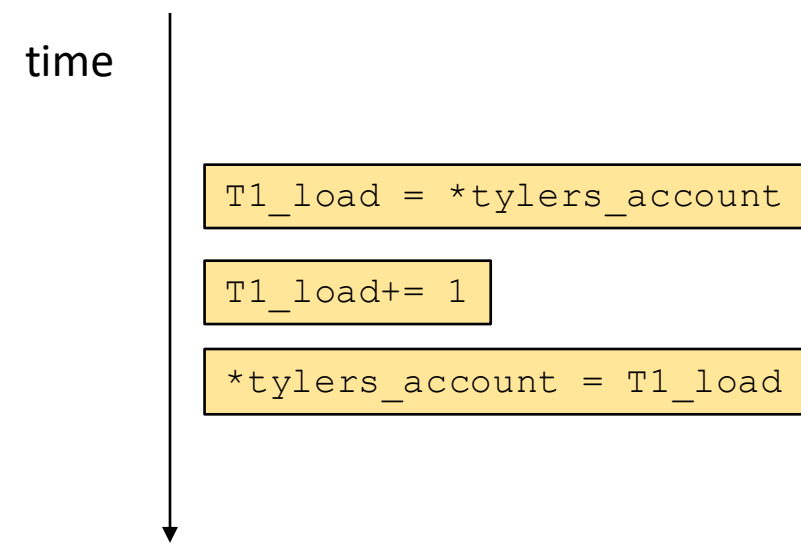
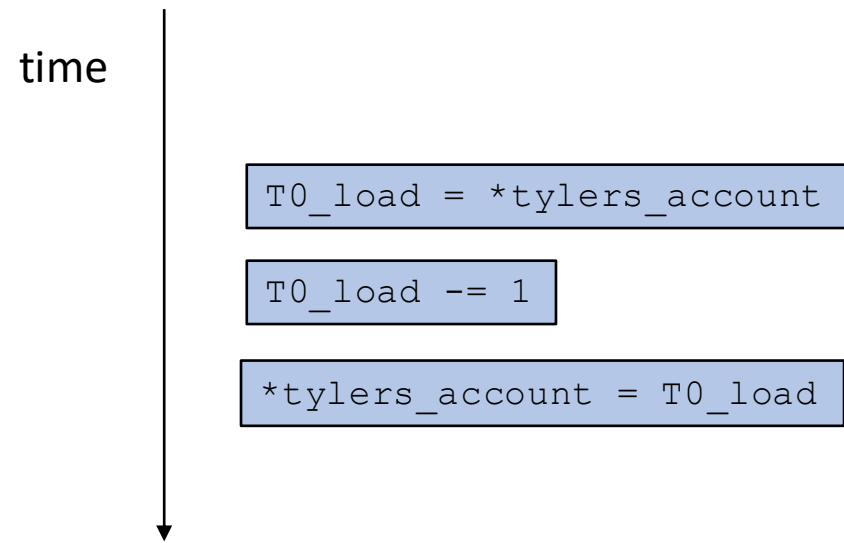


Remember 3 address code...

concurrent execution



time

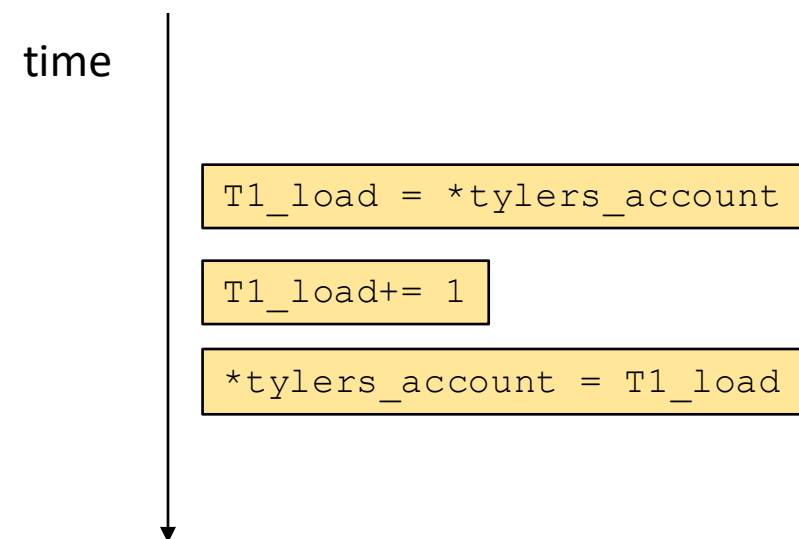
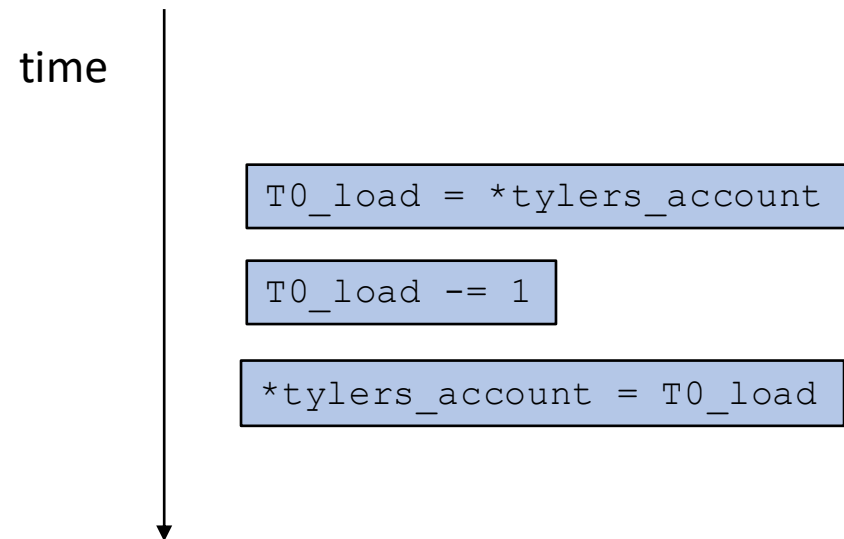


What if we interleave these instructions?

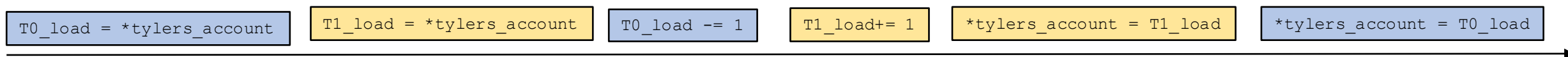
concurrent execution



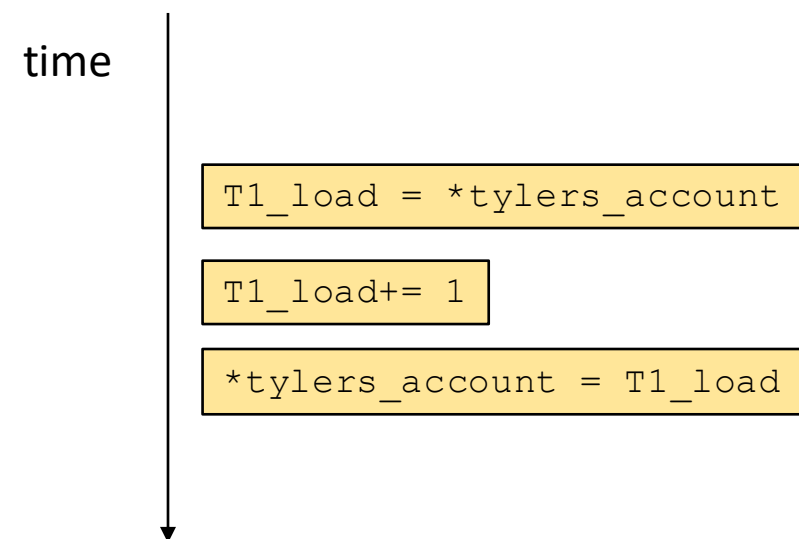
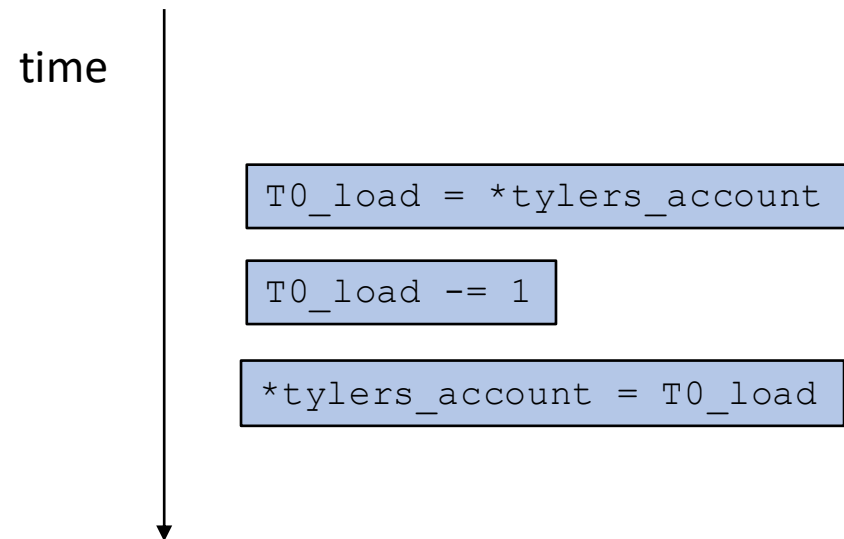
time



concurrent execution

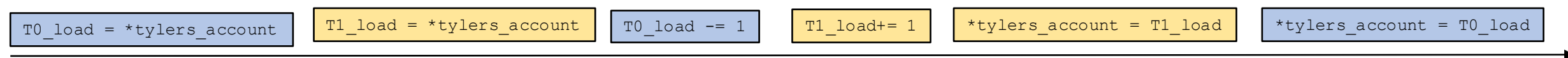


time



*tylers_account has -1 at the
end of this interleaving!*
Atomicity Violation.

concurrent execution



time

What now?

- Data conflicts lead to many different types of issues, not just strange interleavings.
 - Data tearing: one part written from a write and another part from another write.
 - Instruction reorderings: loads can go up and writes can go down.
 - Compiler optimizations: observed from other threads, sequential assumptions of the compiler about a thread are violated.
- Rather than reasoning about data conflicts, we will protect against them using ***synchronization***.

Synchronization

- A scheme where several actors agree on how to safely share a resource during concurrent access.
- Must define what “safely” means.
- Example:
 - Two neighbors sharing a yard between a dog and cat
 - Sharing refrigerator with roommates
 - An account balance that is written to and read from
 - More described in Chapter 1 in text book

Mutexes

- A synchronization object to protect against data conflicts

Simple API:

`lock()`

`unlock()`

- Before a thread accesses the shared memory, it should call `lock()`
- When a thread is finished accessing the shared data, it should call `unlock()`

A thread is a sequential program

Tyler's coffee addiction:

```
tylers_account -= 1;
```

Tyler's employer

```
tylers_account += 1;
```

assume a global mutex object m
protect the account access with the mutex

A thread is a sequential program

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time



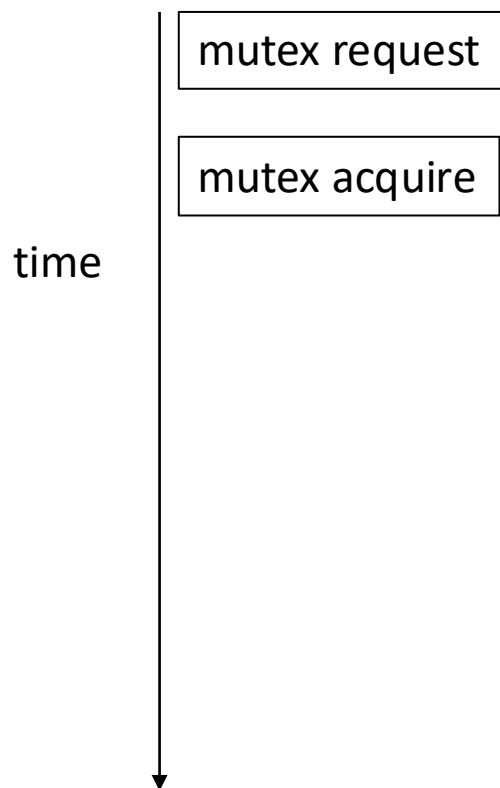
A thread is a sequential program

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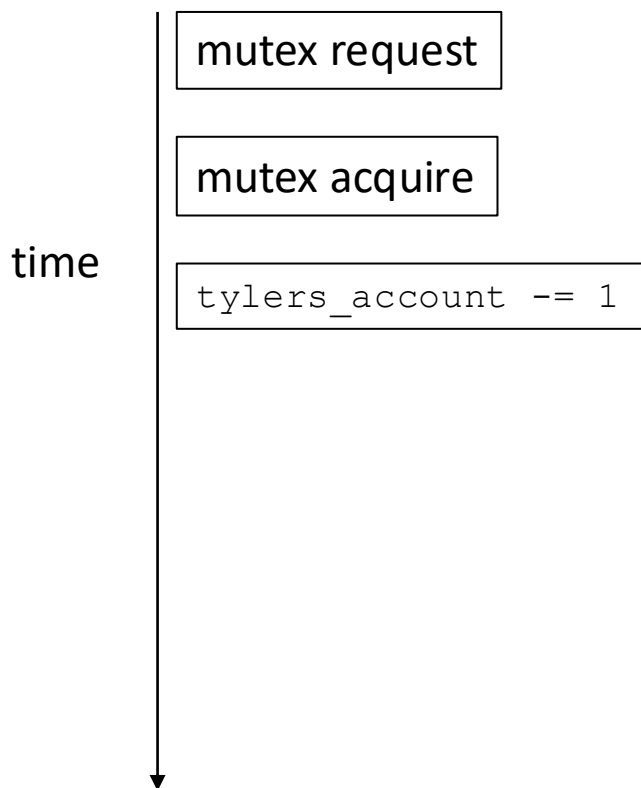
```
m.lock();  
tylers_account += 1;  
m.unlock();
```



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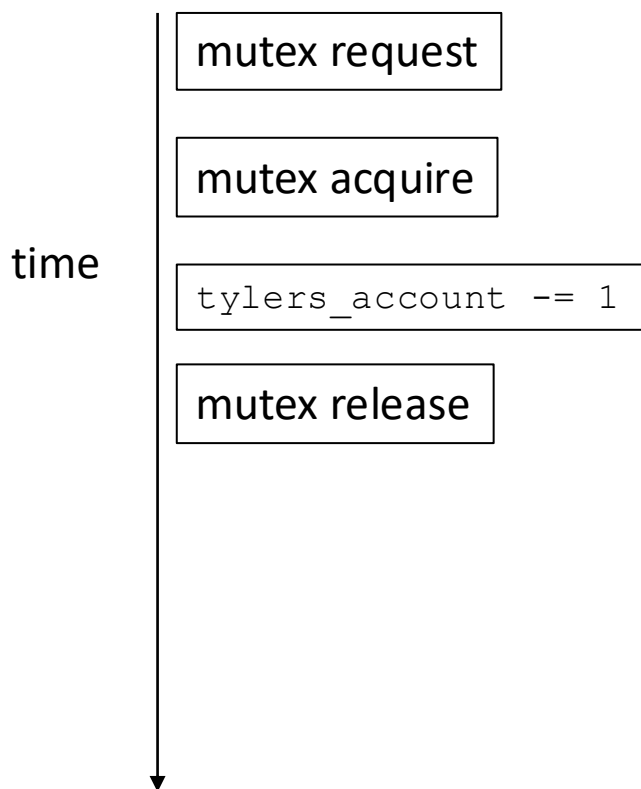
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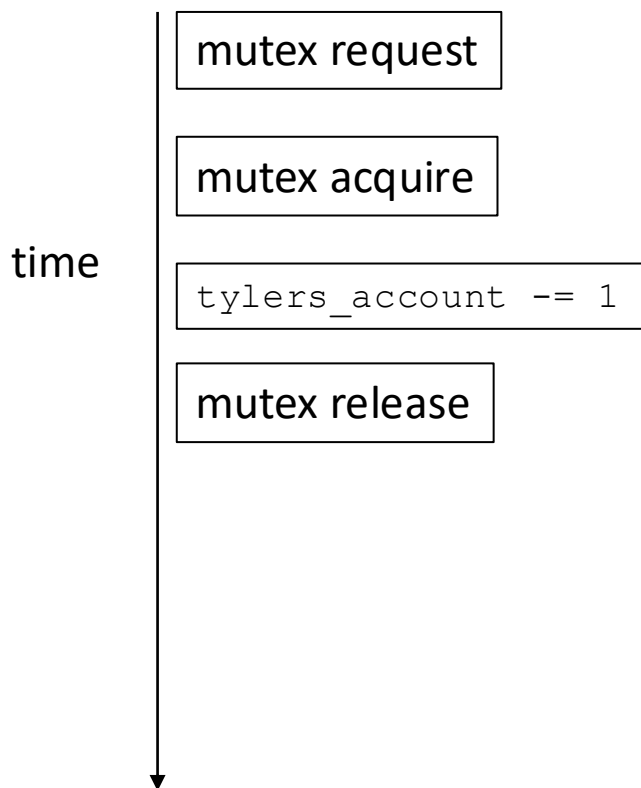
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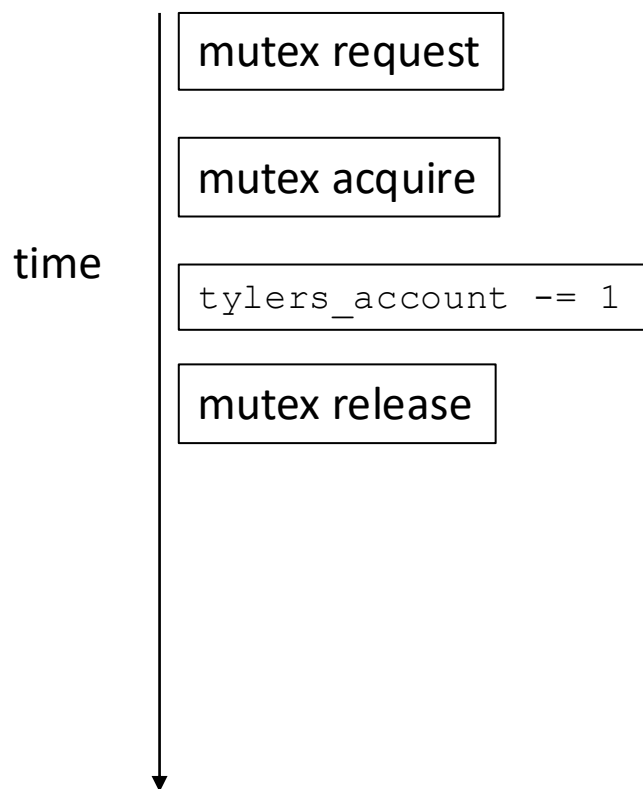
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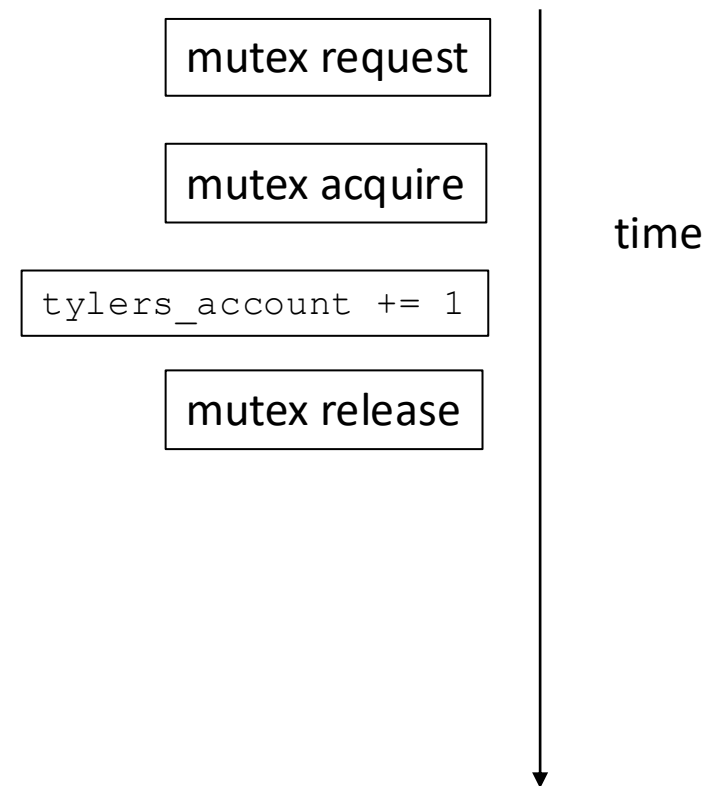
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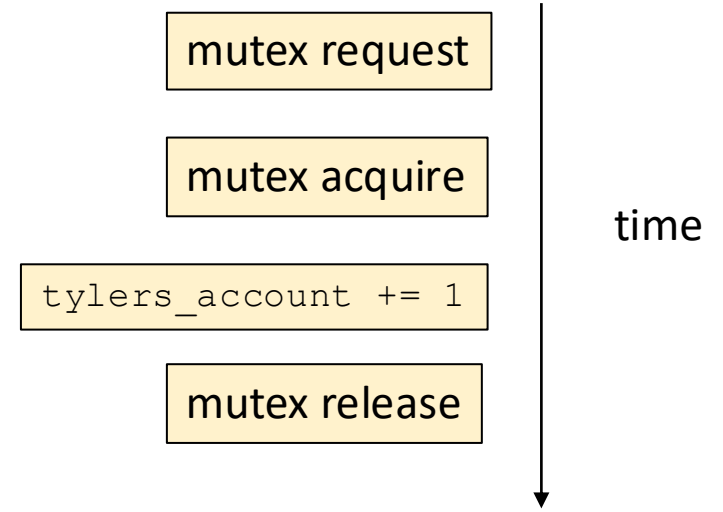
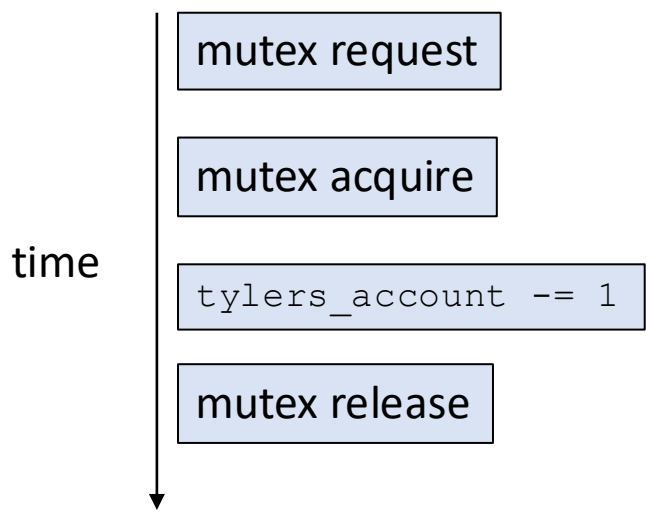
```
m.lock();  
tylers_account -= 1;  
m.unlock();
```



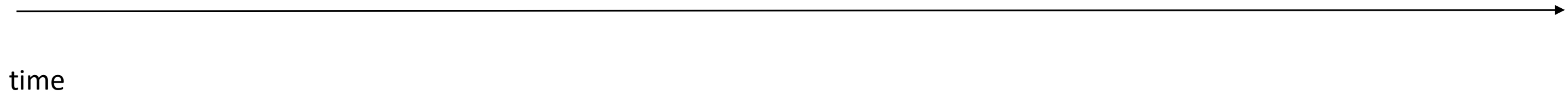
Tyler's employer

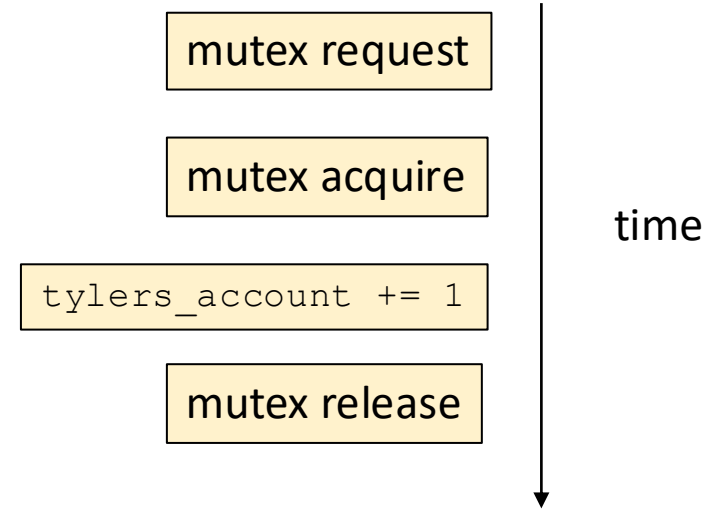
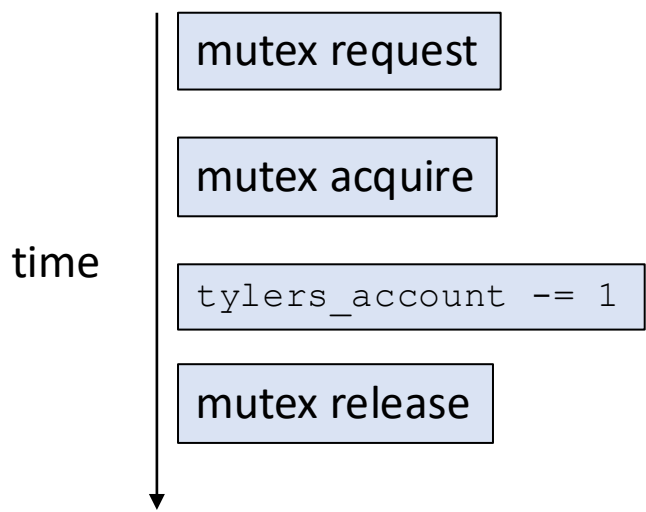
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m.lock();  
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```



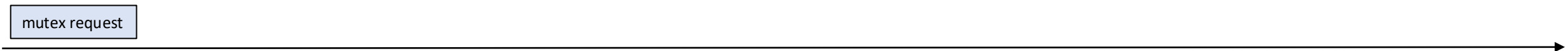


concurrent execution

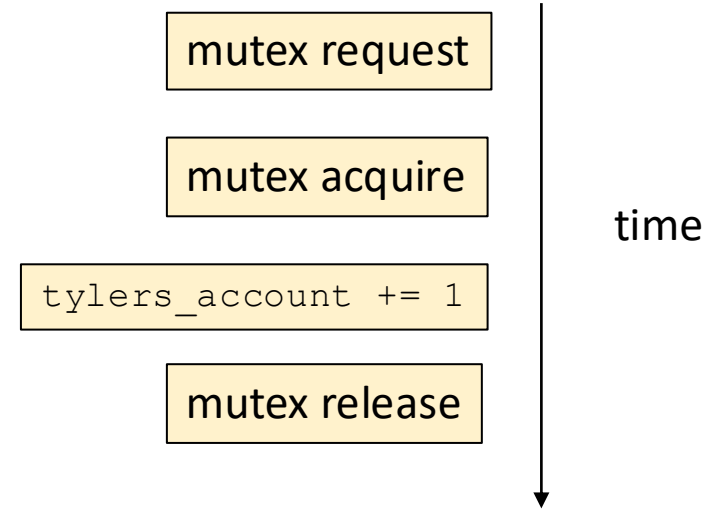
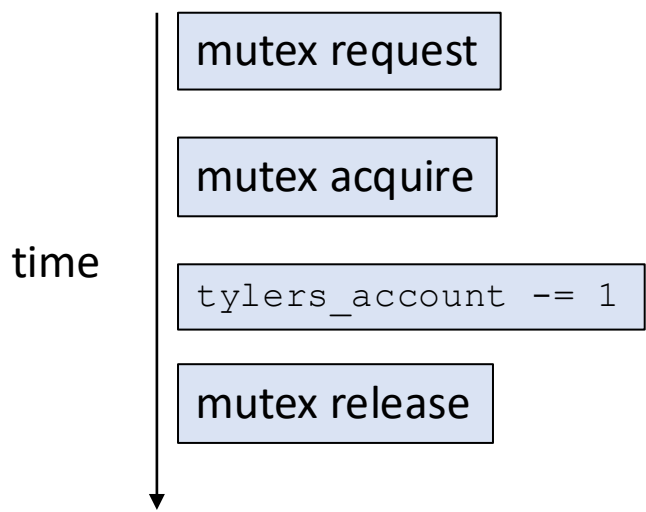




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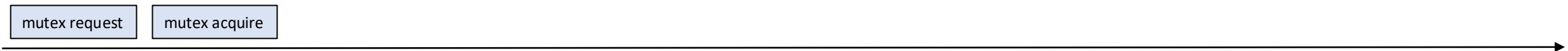


time

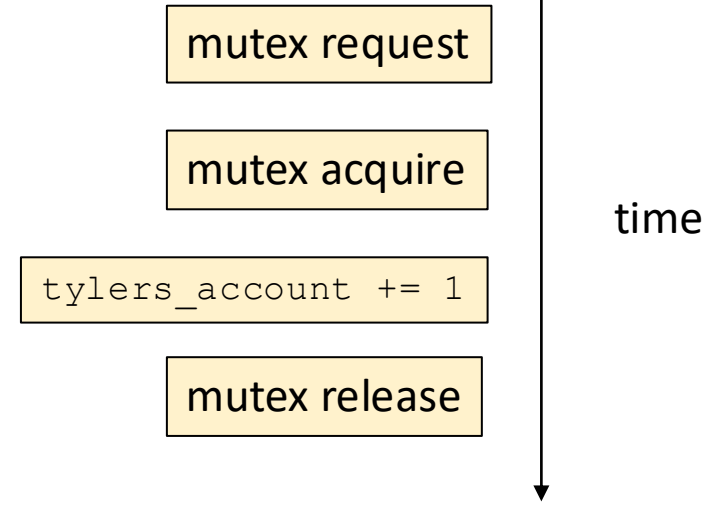
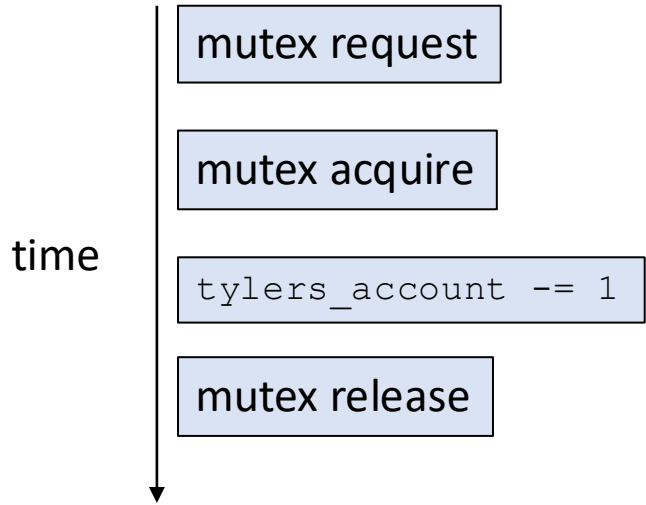


*at this point, thread 0 holds the mutex.
another thread cannot acquire the mutex until thread 0 releases the mutex
also called the **critical section**.*

concurrent execution

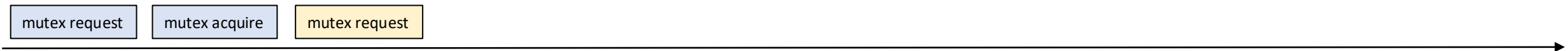


time

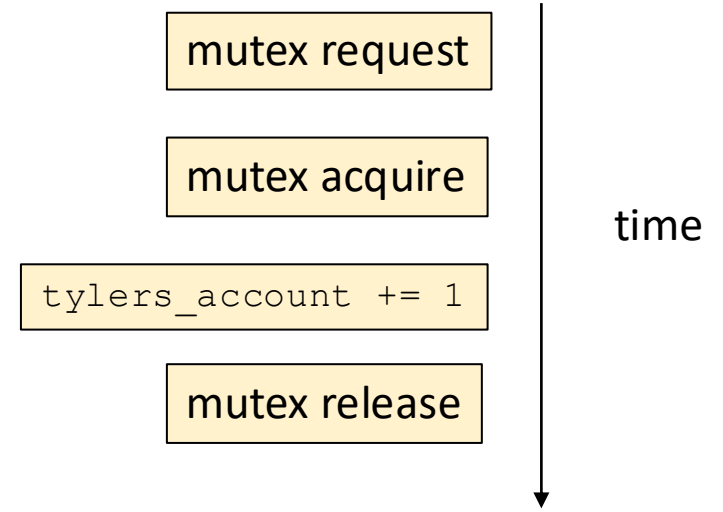
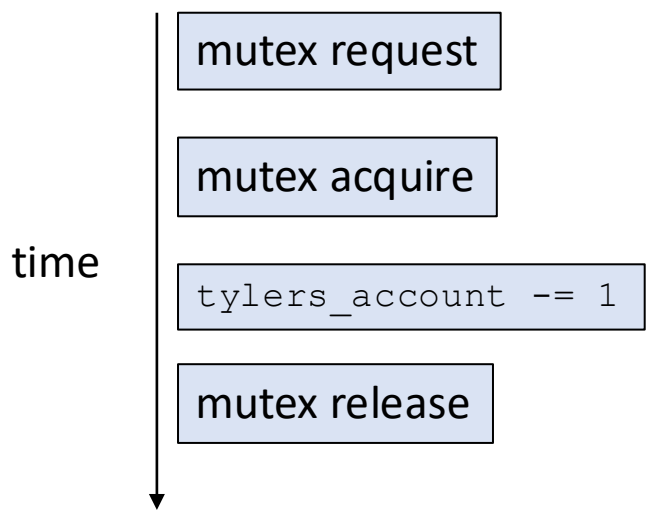


Allowed to request

concurrent execution



time



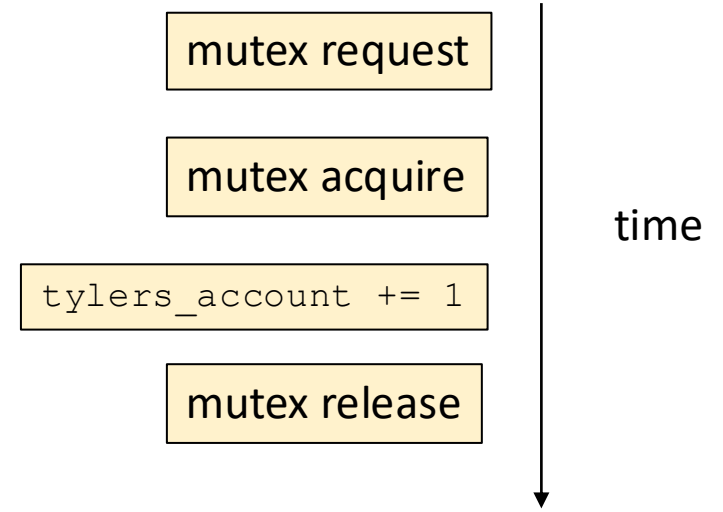
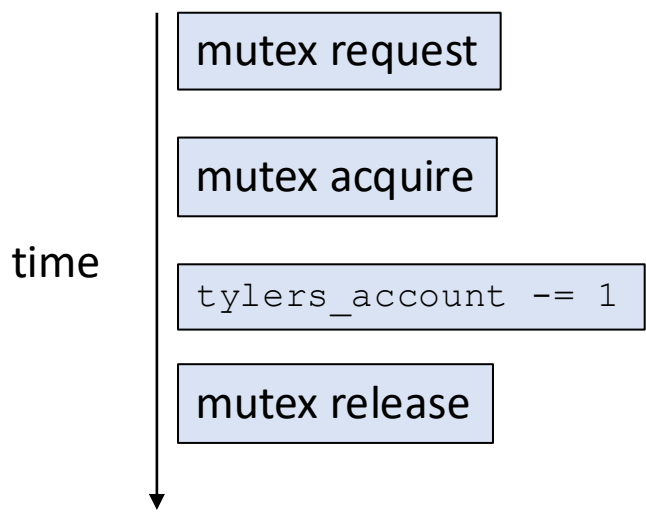
Allowed to request

concurrent execution



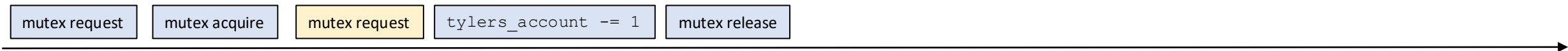
disallowed!

time

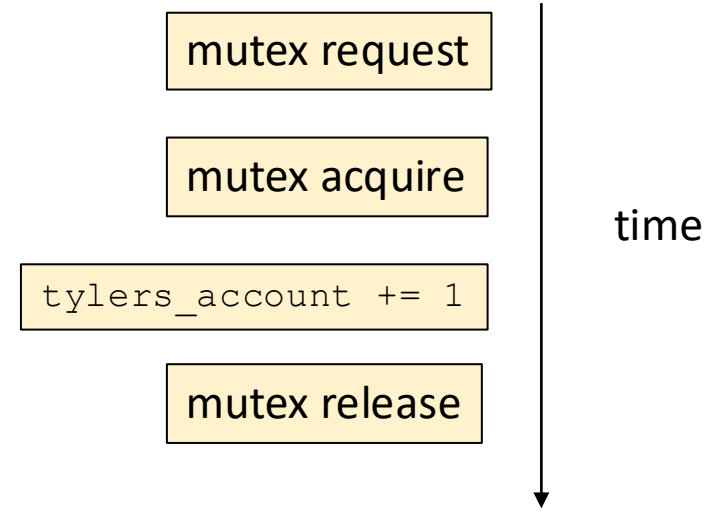
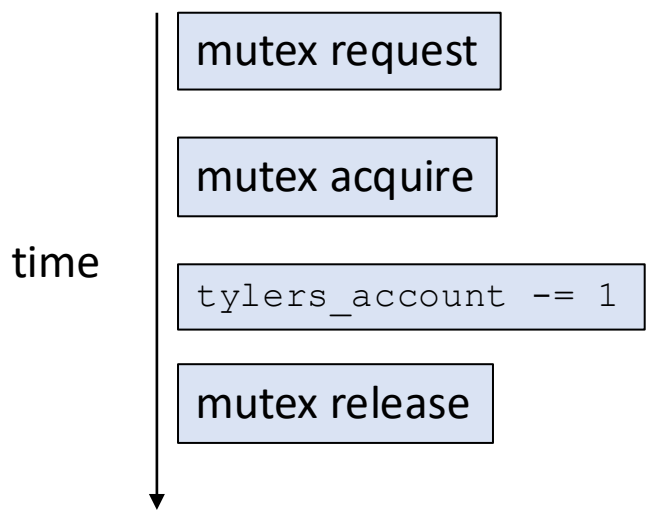


concurrent execution

Thread 0 has released the mutex

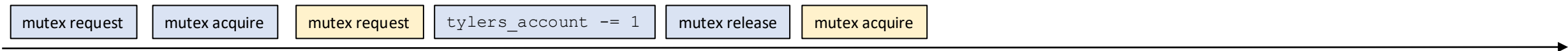


time



concurrent execution

*Thread 1 can take the mutex
and enter the critical section*



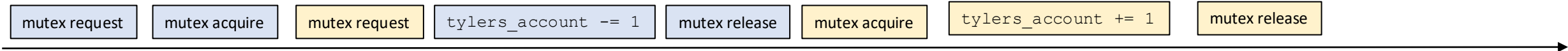
time



A mutex restricts the number of allowed interleavings
Critical section are mutually exclusive: i.e. they cannot interleave

*Thread 1 can take the mutex
and enter the critical section*

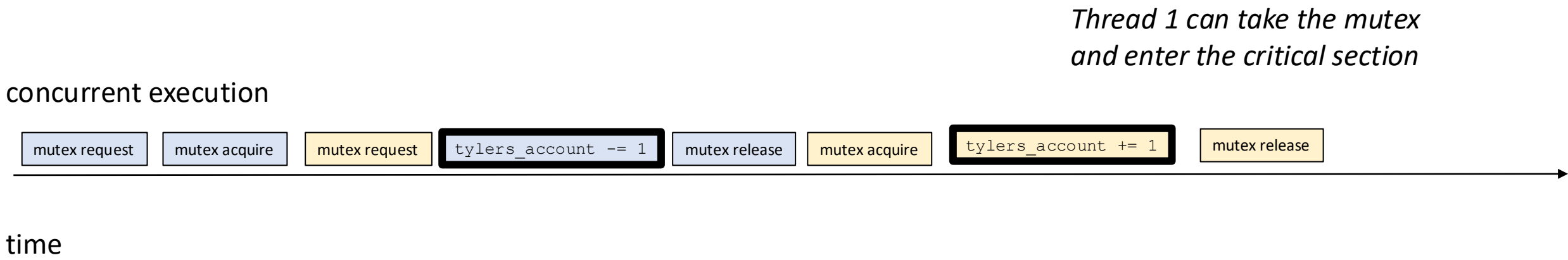
concurrent execution



time



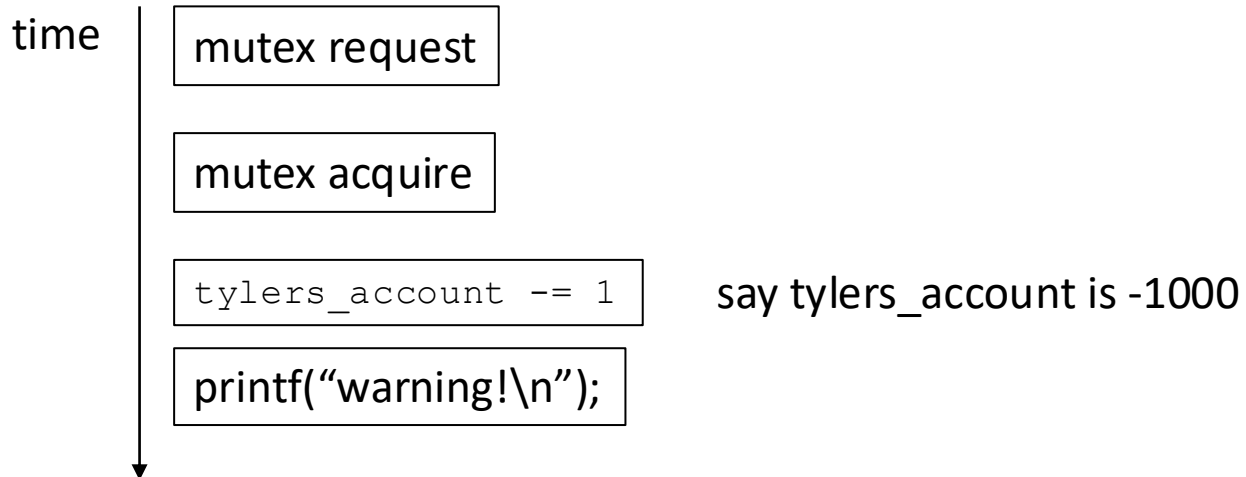
It means we don't have to think about 3 address code



Make sure to unlock your mutex!

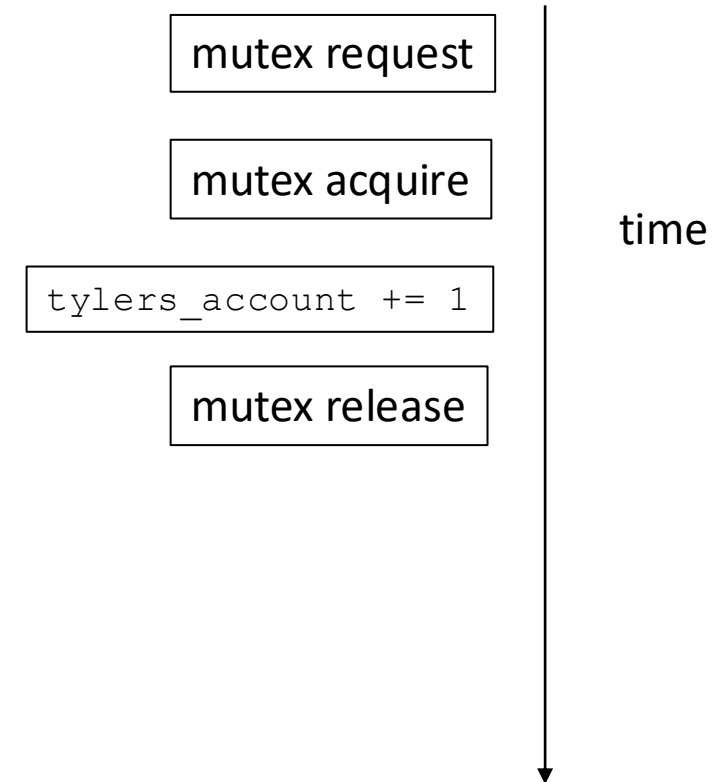
Tyler's coffee addiction:

```
m.lock();
tylers_account -= 1;
if (tylers_account < -100) {
    printf("warning!\n");
    return;
}
m.unlock();
return;
```

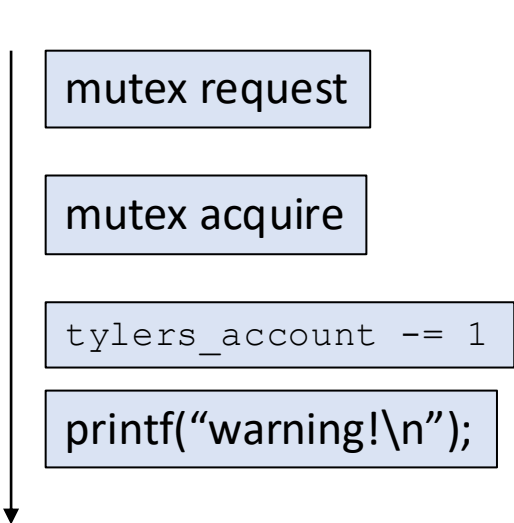


Tyler's employer

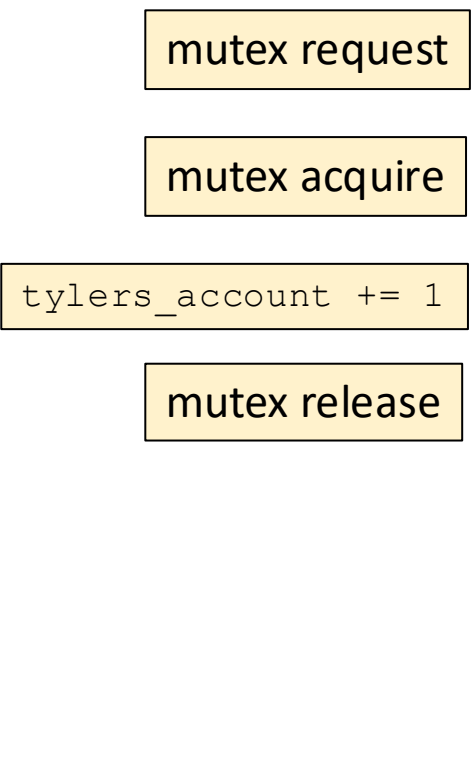
```
m.lock();
tylers_account += 1;
m.unlock();
```



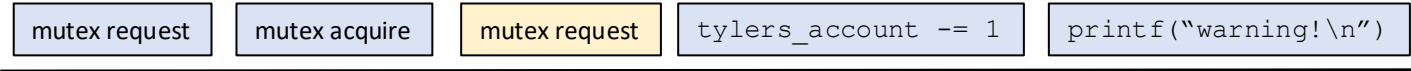
time



time



concurrent execution



Thread 1 is stuck!

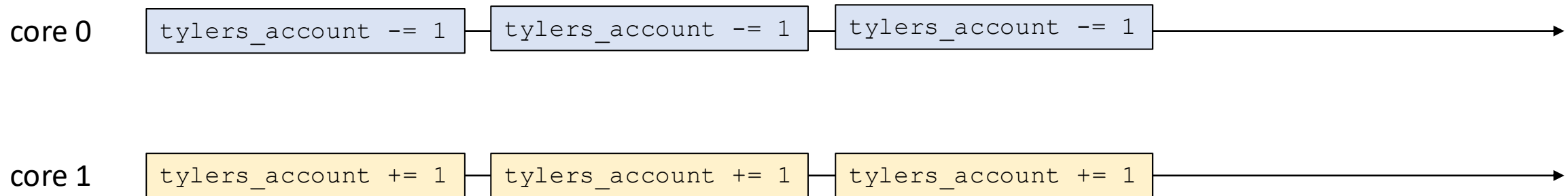
Mutex Performance

- What about timing?
 - Overhead of acquiring/releasing mutex
 - Cache flushing (heavier weight than coherence)
 - Reduces parallelism

Mutex Performance

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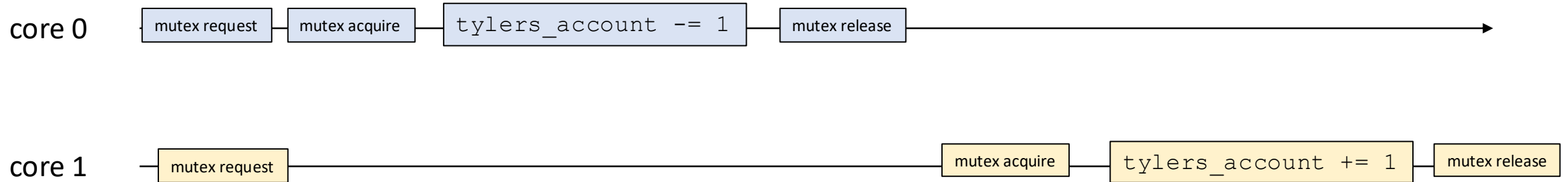
in a parallel system without the mutex



Mutex Performance

- What about timing?
 - Overhead of acquiring/releasing mutex
 - Cache flushing (heavier weight than coherence)
 - Reduces parallelism

*in a parallel system **with** the mutex*



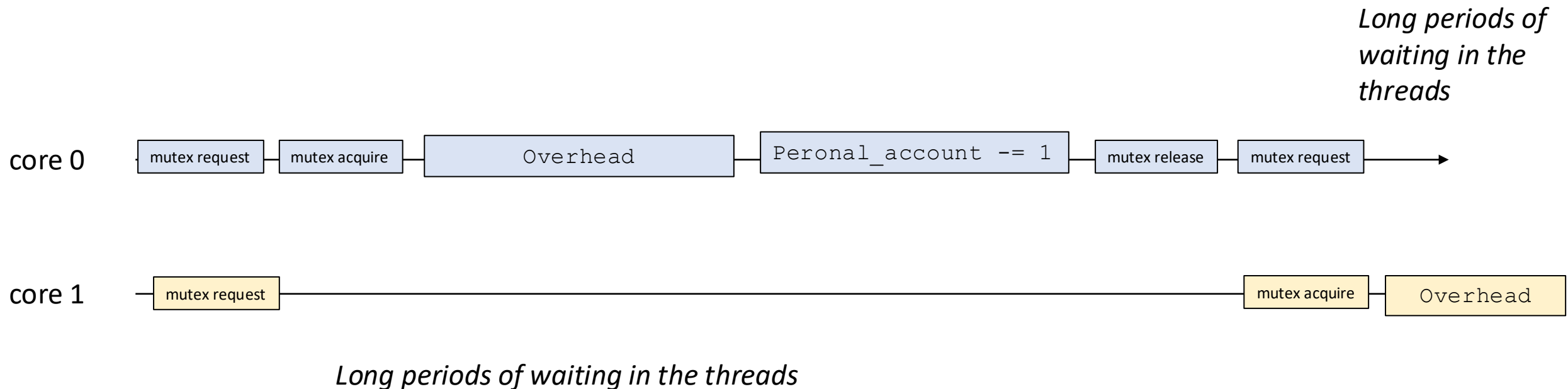
Long periods of waiting in the threads

Mutex Performance

Try to keep mutual exclusion sections small!

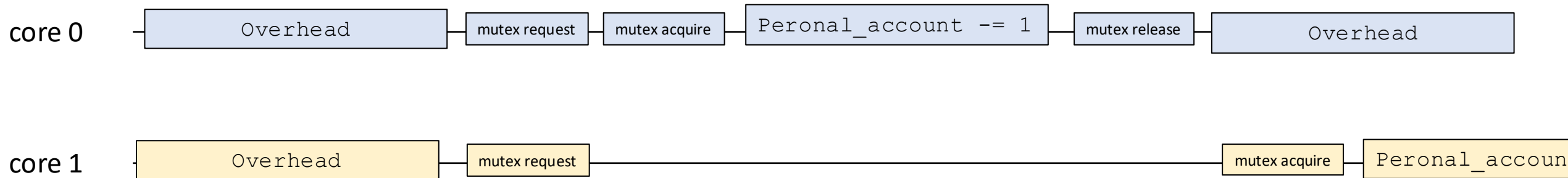
Mutex Performance

Try to keep mutual exclusion sections small! Protect only data conflicts!



Mutex Performance

Try to keep mutual exclusion sections small! Protect only data conflicts!



overlap the overhead (i.e. computation without any data conflicts)

Mutex alternatives?

Other ways to implement accounts?

Atomic Read-modify-write (RMWs): primitive instructions that implement a read event, modify event, and write event indivisibly, i.e. it cannot be interleaved.

```
atomic_fetch_add(atomic_int * addr, int value) {  
    int tmp = *addr; // read  
    tmp += value;    // modify  
    *addr = tmp;     // write  
}
```

other operations: max, min, etc.

Modify these programs to use atomic RMWs

Tyler's coffee addiction:

```
m.lock();  
tylers_account -= 1;  
m.unlock();
```

time



Tyler's employer

```
m.lock();  
tylers_account += 1;  
m.unlock();
```

time



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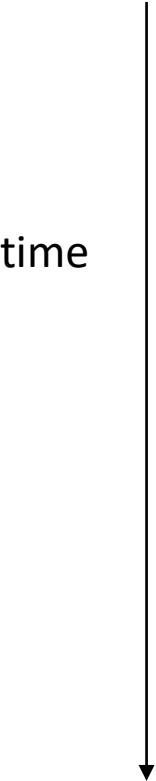
time



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Tyler's coffee addiction:

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tylers_account -= 1;
```



time

Tyler's employer

```
tylers_account += 1;
```



time

Modify these programs to use atomic RMWs

Tyler's coffee addiction:

```
atomic_fetch_add(&tylers_account, -1);
```

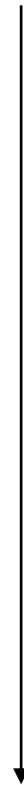
time



Tyler's employer

```
atomic_fetch_add(&tylers_account, 1);
```

time



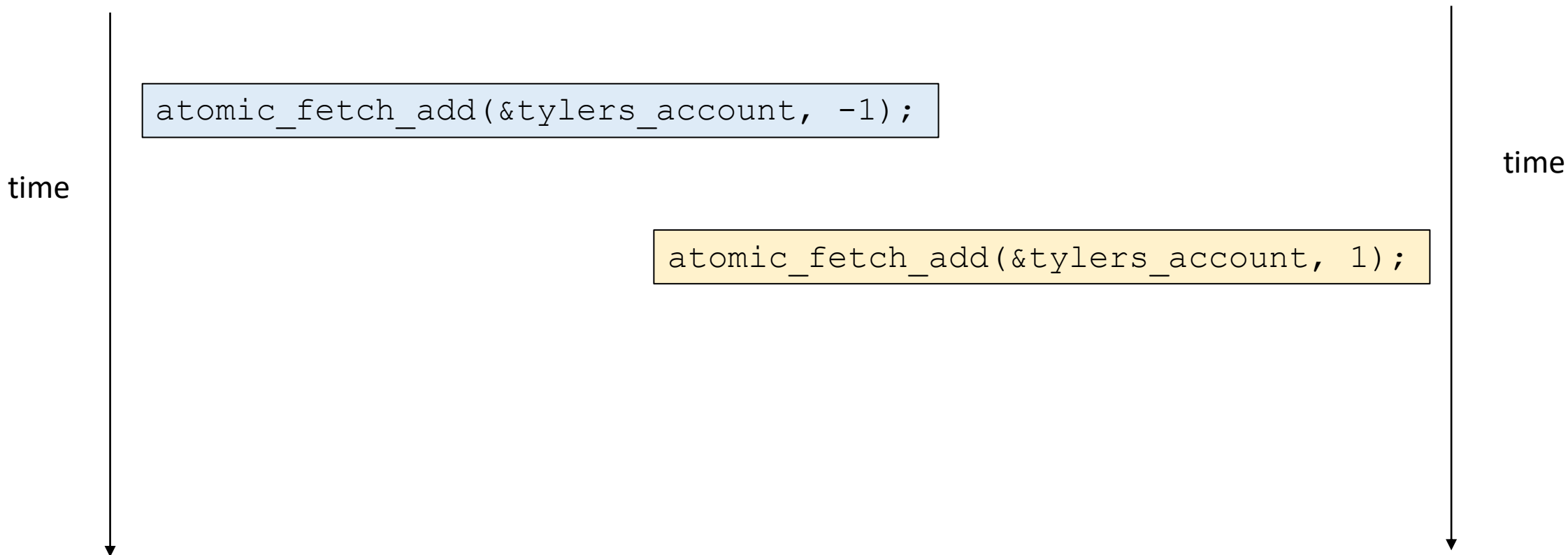
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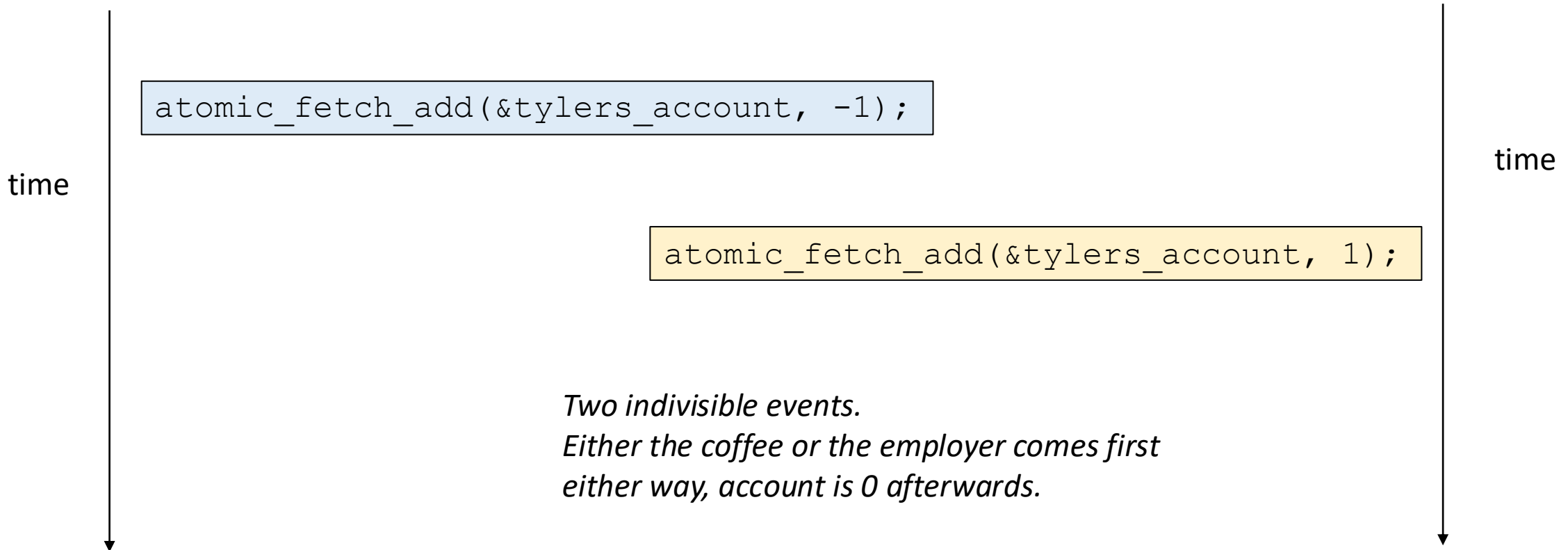
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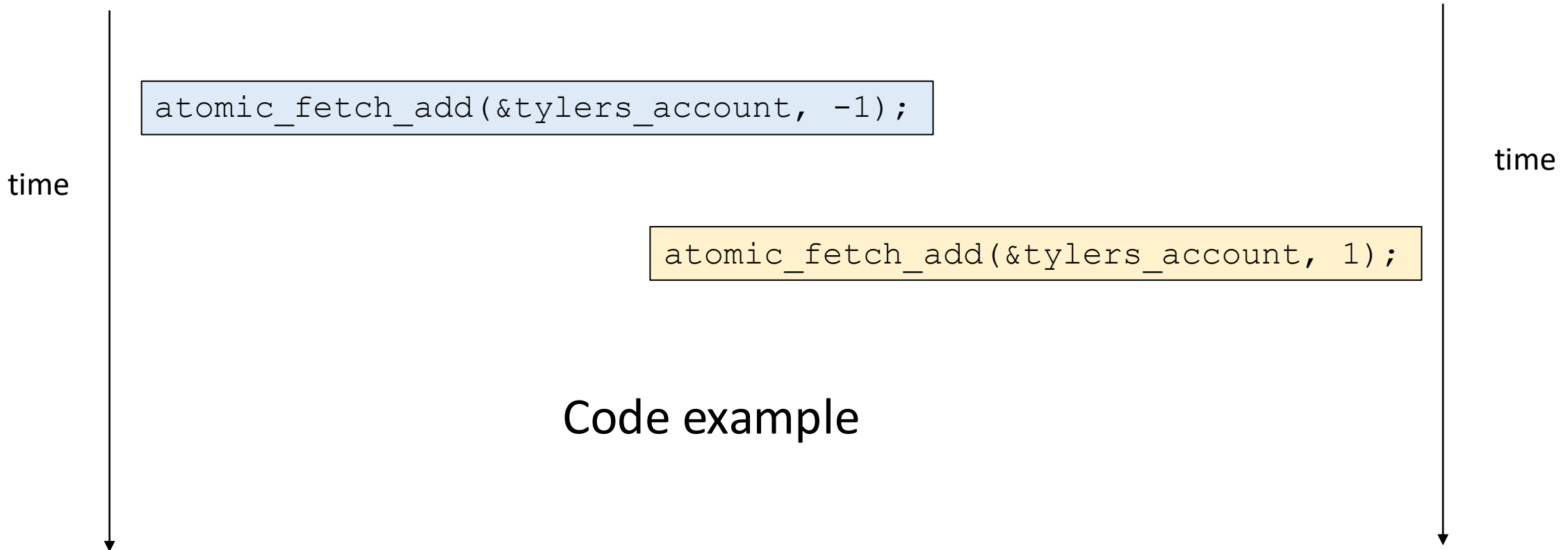
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Tyler's coffee addiction:

```
atomic_fetch_add(&tylers_account, -1);
```

Tyler's employer

```
atomic_fetch_add(&tylers_account, 1);
```



Atomic RMWs

Pros? Cons?

Atomic RMWs

Pros? Cons?

Not all architectures support RMWs (although more common with C++11)

Limits critical section (what if account needs additional updating?)

Multiple mutexes

Lets say I have two accounts:

- Business account
- Personal account
- Need to protect them using a mutex
 - Easy, we can just use the same mutex

Multiple mutexes

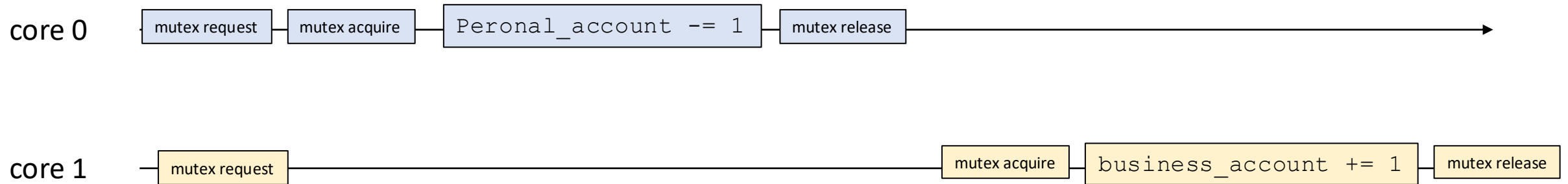
Lets say I have two accounts:

- Business account
 - Personal account
-
- No reason individual accounts can't be accessed in parallel

Multiple mutexes

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- Business account
 - Personal account
-
- No reason individual accounts can't be accessed in parallel



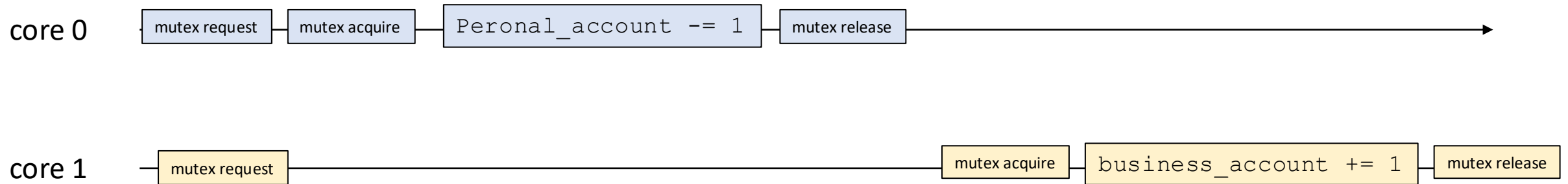
Long periods of waiting in the threads

Multiple mutexes

Mutexes are objects. We can create multiple versions of them to protect different shared data.

MutexP for personal account
MutexB for business account

Critical sections across different mutexes can overlap

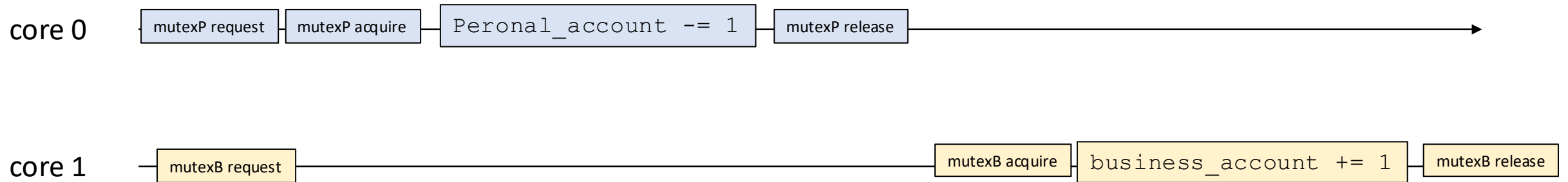


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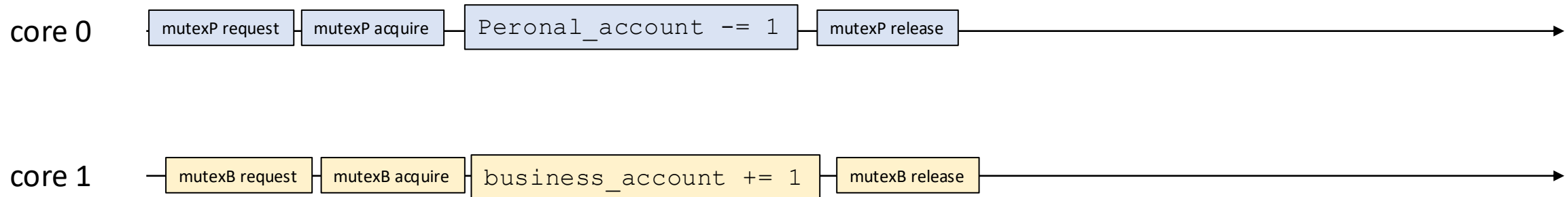


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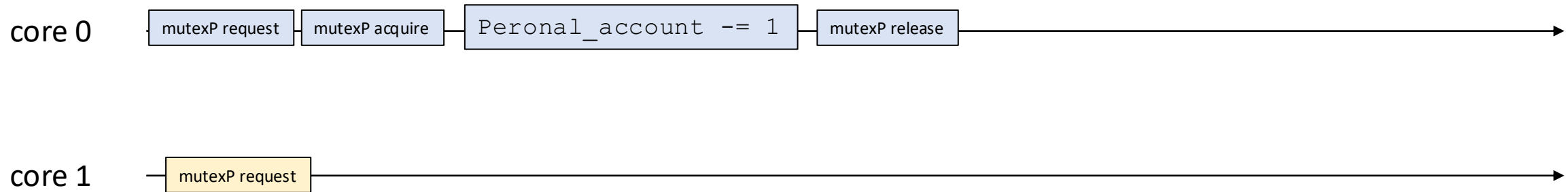


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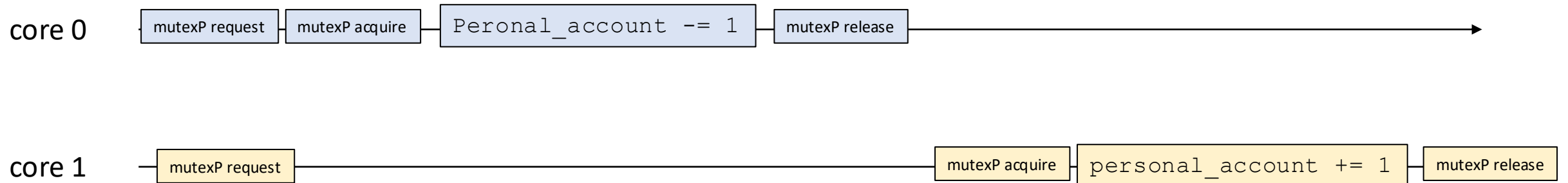


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Mutexes are objects. We can create multiple versions of them to protect different shared data.

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Critical sections across different mutexes can overlap



Managing multiple mutexes

Consider this increasingly elaborate scheme

My accounts start being audited by two agents:

- UCSC
 - IRS
-
- They need to examine the accounts at the same time. They need to acquire both locks

Managing multiple mutexes

Consider this increasingly elaborate scheme

My accounts start being audited by two agents:

- UCSC

- IRS

```
void irs_audit() {  
    for (int i = 0; i < NUM_AUDITS; i++) {  
        tylers_personal_account_mutex.lock();  
        tylers_business_account_mutex.lock();  
  
        AUDIT(tylers_personal_account, tylers_business_account);  
  
        tylers_personal_account_mutex.unlock();  
        tylers_business_account_mutex.unlock();  
    }  
}
```

Multiple mutexes

- Our program deadlocked! What happened?

```
void ucsc_audit() {  
    for (int i = 0; i < NUM_AUDITS; i++) {  
        tylers_business_account_mutex.lock();  
        tylers_personal_account_mutex.lock();  
  
        AUDIT(tylers_personal_account, tylers_business_account);  
  
        tylers_personal_account_mutex.unlock();  
        tylers_business_account_mutex.unlock();  
    }  
}
```

Multiple mutexes

- Our program deadlocked! What happened?



Multiple mutexes

- Our program deadlocked! What happened?



Multiple mutexes

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Multiple mutexes

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Multiple mutexes

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Multiple mutexes

- Our program deadlocked! What happened?



Multiple mutexes

- Our program deadlocked! What happened?

IRS has the personal mutex and won't release it until it acquires the business mutex.

UCSC has the business mutex and won't release it until it acquires the personal mutex.

This is called a deadlock!



Multiple mutexes

- Our program deadlocked! What happened?
- Fix: Acquire mutexes in the same order
- Proof sketch by contradiction
 - Thread 0 is holding mutex X waiting for mutex Y
 - Thread 1 is holding mutex Y waiting for mutex X

Assume the order that you acquire mutexes is X then Y

Thread 1 cannot hold mutex Y without holding mutex X.

Thread 0 cannot hold mutex X because thread 1 is holding mutex X.

Programming with mutexes can be HARD!

make sure all data conflicts are protected with a mutex

keep critical sections small

balance between having many mutexes (provides performance) but gives the potential for deadlocks

Next time

- Implementing Mutexes